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Reference signal patterns based on relative speed between a transmitter and receiver

Abstract

Methods, systems, and devices for wireless communications (e.g., vehicle to everything systems) are described relating to an adaptive design of demodulation reference signals (DMRS) density based on user equipment (UE) velocity. A UE may send assistance information to a transmitting UE to help identify a DMRS pattern based on the relative speed between the two UEs. Also, the transmitter may determine the adaptive DMRS pattern based on its speed without information of the receiver's speed. The transmitter may indicate the adaptive DMRS pattern in control information to the receiver. A base station may receive assistance information, and the base station may determine the adaptive DMRS pattern to be used by the transmitting UE based on the received assistance information and then indicate the adaptive DMRS pattern to the transmitting UE. A UE may determine an adaptive DMRS pattern to use based on feedback from the receiving UE.

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Background/Summary

CROSS REFERENCE (1) The present Application for Patent is a Divisional of U.S. patent application Ser. No. 16/843,797 by MALLADI et al., entitled “REFERENCE SIGNAL PATTERNS BASED ON RELATIVE SPEED BETWEEN A TRANSMITTER AND RECEIVER” filed Apr. 8, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/833,635 by MALLADI et al., entitled “REFERENCE SIGNAL PATTERNS BASED ON RELATIVE SPEED BETWEEN A TRANSMITTER AND RECEIVER,” filed Apr. 12, 2019, each of which are assigned to the assignee hereof, and each of which are expressly incorporated by reference in its entirety herein.

INTRODUCTION
(1) The following relates generally to wireless communications, and more specifically to reference signal patterns.
(2) Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such multiple-access systems include fourth generation (4G) systems such as Long Term Evolution (LTE) systems, LTE-Advanced (LTE-A) systems, or LTE-A Pro systems, and fifth generation (5G) systems which may be referred to as New Radio (NR) systems. These systems may employ technologies such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal frequency division multiple access (OFDMA), or discrete

Fourier transform spread orthogonal frequency division multiplexing (DFT-S-OFDM). A wireless multiple-access communications system may include a number of base stations or network access nodes, each simultaneously supporting communication for multiple communication devices, which may be otherwise known as user equipment (UE).

SUMMARY

(3) A method of wireless communications at a first wireless device is described. The method may include transmitting assistance information to a second wireless device and determining an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern based on the assistance information.

(4) An apparatus for wireless communications at a first wireless device is described. The apparatus may include a processor and memory coupled to the processor. The processor and memory may be configured to cause the apparatus to transmit assistance information to a second wireless device and determine an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern based on the assistance information.

(5) Another apparatus for wireless communications at a first wireless device is described. The apparatus may include means for transmitting assistance information to a second wireless device and determining an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern based on the assistance information.

(6) A non-transitory computer-readable medium storing code for wireless communications at a first wireless device is described. The code may include instructions executable by a processor to transmit assistance information to a second wireless device and determine an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern based on the assistance information.

(7) A method of wireless communications at a first wireless device is described. The method may include transmitting assistance information to a second wireless device, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof, and identifying a reference signal pattern of a reference signal for demodulating data, the reference signal pattern based on the assistance information.

(8) An apparatus for wireless communications at a first wireless device is described. The apparatus may include a processor and memory coupled to the processor. The processor and memory may be configured to cause the apparatus to transmit assistance information to a second wireless device, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof, and identify a reference signal pattern of a reference signal for demodulating data, the reference signal pattern based on the assistance information.

(9) Another apparatus for wireless communications at a first wireless device is described. The apparatus may include means for transmitting assistance information to a second wireless device, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof, and identifying a reference signal pattern of a reference signal for demodulating data, the reference signal pattern based on the assistance information.

(10) A non-transitory computer-readable medium storing code for wireless communications at a first wireless device is described. The code may include instructions executable by a processor to transmit assistance information to a second wireless device, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof, and identify a reference signal pattern of a reference signal for demodulating data, the reference signal pattern based on the assistance information.

(11) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining a speed of the first wireless device, and transmitting, as part of the assistance information, a speed value indicating the speed of the first wireless device.

(12) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining a movement direction of the first wireless device, and transmitting, as part of the assistance information, a direction value indicating the movement direction of the first wireless device.
(13) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining an absolute movement direction of the first wireless device, and identifying the direction value from a direction index based on the absolute movement direction, the direction index including a map between one or more absolute movement directions and respective direction values.
(14) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, each of the respective direction values may be based on an angular offset from a cardinal direction or an intercardinal direction. Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining an absolute speed of the first wireless device, and identifying the speed value from a speed index based on the absolute speed, the speed index including a map between respective speed values and one or more absolute speeds, or a range of absolute speeds, or a combination thereof. In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the speed index may be from a set of speed indices.
(15) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a configuration of the speed index, where identifying the speed value may be based on the configuration. Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving an indication of the configuration of the speed index.
(16) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining a location of the first wireless device, and transmitting, as part of the assistance information, a location value indicating the location of the first wireless device.
(17) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a zone identifier from a set of zone identifiers based on the location of the first wireless device, each of the set of zone identifiers being associated with a respective geographic area, where the location value includes the zone identifier.
(18) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a relative speed between the first wireless device and the second wireless device based on the assistance information, where the adaptive reference signal pattern may be determined based on the relative speed.
(19) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, identifying the relative speed may include operations, features, means, or instructions for receiving, from the second wireless device, an indication of a speed of the second wireless device, or a velocity of the second wireless device, or a location of the second wireless device, or a combination thereof, and determining the relative speed based on the received indication.
(20) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, determining the relative speed may include operations, features, means, or instructions for determining the speed of the first wireless device, and calculating the relative speed based on the speed of the first wireless device and the speed of the second wireless device.
(21) In some examples of the method, apparatuses, and non-transitory computer-readable medium

described herein, determining the relative speed may include operations, features, means, or instructions for determining a velocity of the first wireless device, and calculating the relative speed based on the velocity of the first wireless device and the velocity of the second wireless device.

(22) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, determining the relative speed may include operations, features, means, or instructions for determining, based on the location of the second wireless device, one or more zones the second wireless device was located in during a time period, where the received indication includes one or more zone identifiers corresponding to respective zones, calculating the speed of the second wireless device based on the one or more zones the second wireless device was located in during the time period, and calculating the relative speed based on the speed of the first wireless device and the speed of the second wireless device.

(23) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting, as part of the assistance information, an indication of the relative speed. Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining the adaptive reference signal pattern based on the relative speed, and transmitting, as part of the assistance information, an indication of the adaptive reference signal pattern.

(24) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, determining the adaptive reference signal pattern may include operations, features, means, or instructions for receiving, from the second wireless device, an indication of the adaptive reference signal pattern, and determining the adaptive reference signal pattern based on the indication.

(25) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, determining the adaptive reference signal pattern may include operations, features, means, or instructions for receiving, from a set of wireless devices, respective indications of one or more reference signal patterns, and determining the adaptive reference signal pattern based on the one or more reference signal patterns.

(26) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, determining the adaptive reference signal pattern may include operations, features, means, or instructions for selecting the adaptive reference signal pattern from one or more reference signal patterns, each of the one or more reference signal patterns corresponding to a respective relative speed, or a range of relative speeds, or a combination thereof.

(27) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a configuration of a map between the each of the one or more reference signal patterns and the respective relative speed, or the range of relative speeds, or a combination thereof.

(28) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, determining the adaptive reference signal pattern may include operations, features, means, or instructions for identifying a set of communication links with other wireless devices, each communication link of the set of communication links being associated with a relative speed between the first wireless device and a respective wireless device, and determining the adaptive reference signal pattern based on the relative speeds of the set of communication links.

(29) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a highest relative speed from the relative speeds of the set of communication links, where the adaptive reference signal pattern may be based on the highest relative speed.

(30) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining an

error rate threshold for transmitting the data over the set of communication links, where the adaptive reference signal pattern may be based on the relative speeds of the set of communication links and the error rate threshold.
(31) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, determining the adaptive reference signal pattern may include operations, features, means, or instructions for identifying a subcarrier spacing for communicating with the second wireless device, and determining the adaptive reference signal pattern based on the subcarrier spacing and a relative speed between the first wireless device and the second wireless device.
(32) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the adaptive reference signal pattern may be based on a map between the one or more reference signal patterns and a respective relative speed, or a range of relative speeds, or a combination thereof, and where the adaptive reference signal pattern may be determined based on the map and the subcarrier spacing.
(33) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving first assistance information and second assistance information from the second wireless device, each of the first assistance information and the second assistance information being associated with a respective speed of the second wireless device, or a respective location of the second wireless device, or a combination thereof, and storing the first assistance information and the second assistance information at the first wireless device.
(34) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a difference between the first assistance information and the second assistance information, and determining the adaptive reference signal pattern based on the difference. Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for identifying a periodicity for transmitting the assistance information, where the assistance information may be transmitted in accordance with the periodicity.
(35) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for generating first assistance information and second assistance information for the first wireless device, each of the first assistance information and the second assistance information being associated with a respective speed of the first wireless device, or a respective location of the first wireless device, or a combination thereof, identifying a difference between the first assistance information and the second assistance information, and transmitting the assistance information based on identifying the difference.
(36) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving a request to transmit the assistance information, where the assistance information may be transmitted in response to the received request.
(37) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting the data and the reference signal to the second wireless device, the reference signal being transmitted in accordance with the adaptive reference signal pattern. Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for receiving the data and the reference signal from the second wireless device, the reference signal being received in accordance with the adaptive reference signal pattern.
(38) In some examples of the method, apparatuses, and non-transitory computer-readable medium

described herein, transmitting the assistance information may include operations, features, means, or instructions for transmitting the assistance information as part of sidelink control information, or as part of a medium access control (MAC) control element, or over a sidelink data channel, or over a sidelink shared channel, or over a feedback channel, or a combination thereof.
(39) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the adaptive reference signal pattern indicates one or more symbol periods during which the reference signal may be transmitted and a gap between each of the one or more symbol periods.
(40) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the adaptive reference signal pattern may be from a set of reference signal patterns, each reference signal pattern of the set including a different number of symbol periods within a slot that include the reference signal.
(41) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the adaptive reference signal pattern may be from a set of reference signal patterns, each reference signal pattern of the set including a different gap between symbol periods within a slot that include the reference signal.
(42) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for communicating with the second wireless device using sidelink communications, where the first wireless device includes a first UE and the second wireless device includes a second UE. In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the sidelink communications may be vehicle to everything communications.
(43) A method of wireless communications at a first wireless device is described. The method may include determining a speed of the first wireless device, identifying a subcarrier spacing for communicating with one or more other wireless devices, determining, based on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data, and transmitting the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(44) An apparatus for wireless communications at a first wireless device is described. The apparatus may include a processor and memory coupled to the processor. The processor and memory may be configured to cause the apparatus to determine a speed of the first wireless device, identify a subcarrier spacing for communicating with one or more other wireless devices, determine, based on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data, and transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(45) Another apparatus for wireless communications at a first wireless device is described. The apparatus may include means for determining a speed of the first wireless device, identifying a subcarrier spacing for communicating with one or more other wireless devices, determining, based on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data, and transmitting the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(46) A non-transitory computer-readable medium storing code for wireless communications at a first wireless device is described. The code may include instructions executable by a processor to determine a speed of the first wireless device, identify a subcarrier spacing for communicating with one or more other wireless devices, determine, based on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data, and transmit the data and the reference signal to the one or more other wireless devices, the reference

signal being transmitted in accordance with the reference signal pattern.
(47) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, transmitting the data and indication of the reference signal pattern may include operations, features, means, or instructions for broadcasting the data and the reference signal to the one or more other wireless devices.
(48) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for transmitting an indication of the reference signal pattern to the one or more other wireless devices. In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the indication may be transmitted within control information to the one or more other wireless devices.
(49) A method of wireless communications at a first wireless device is described. The method may include transmitting assistance information to a base station, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof, receiving, from the base station, an indication of an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern being based on the assistance information, and transmitting the data and the reference signal to a second wireless device, the reference signal being transmitted in accordance with the adaptive reference signal pattern.
(50) An apparatus for wireless communications at a first wireless device is described. The apparatus may include a processor and memory coupled to the processor. The processor and memory may be configured to cause the apparatus to transmit assistance information to a base station, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof, receive, from the base station, an indication of an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern being based on the assistance information, and transmit the data and the reference signal to a second wireless device, the reference signal being transmitted in accordance with the adaptive reference signal pattern.
(51) Another apparatus for wireless communications at a first wireless device is described. The apparatus may include means for transmitting assistance information to a base station, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof, receiving, from the base station, an indication of an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern being based on the assistance information, and transmitting the data and the reference signal to a second wireless device, the reference signal being transmitted in accordance with the adaptive reference signal pattern.
(52) A non-transitory computer-readable medium storing code for wireless communications at a first wireless device is described. The code may include instructions executable by a processor to transmit assistance information to a base station, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof, receive, from the base station, an indication of an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern being based on the assistance information, and transmit the data and the reference signal to a second wireless device, the reference signal being transmitted in accordance with the adaptive reference signal pattern.
(53) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the adaptive reference signal pattern may be from one or more reference signal patterns, each of the one or more reference signal patterns corresponding to a respective speed, or a range of speeds, or a combination thereof.
(54) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for operating

under a mode for sidelink communications, where the first wireless device includes a first UE and the second wireless device includes a second UE.
(55) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the adaptive reference signal pattern includes one or more symbol periods during which the reference signal may be transmitted and a gap between each of the one or more symbol periods.
(56) A method of wireless communications at a first wireless device is described. The method may include receiving, from one or more other wireless devices, feedback information in response to one or more previous data transmissions to the one or more other wireless devices, determining a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the feedback information, and transmitting the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(57) An apparatus for wireless communications at a first wireless device is described. The apparatus may include a processor and memory coupled to the processor. The processor and memory may be configured to cause the apparatus to receive, from one or more other wireless devices, feedback information in response to one or more previous data transmissions to the one or more other wireless devices, determine a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the feedback information, and transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(58) Another apparatus for wireless communications at a first wireless device is described. The apparatus may include means for receiving, from one or more other wireless devices, feedback information in response to one or more previous data transmissions to the one or more other wireless devices, determining a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the feedback information, and transmitting the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(59) A non-transitory computer-readable medium storing code for wireless communications at a first wireless device is described. The code may include instructions executable by a processor to receive, from one or more other wireless devices, feedback information in response to one or more previous data transmissions to the one or more other wireless devices, determine a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the feedback information, and transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(60) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, determining the reference signal pattern may include operations, features, means, or instructions for computing an error rate over a time period based on the feedback information, and determining the reference signal pattern based on the error rate satisfying an error rate threshold.
(61) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the error rate includes a block error rate or a packet error rate of the one or more previous data transmissions.
(62) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, determining the reference signal pattern may include operations, features, means, or instructions for measuring a received power of the feedback information, and determining the reference signal pattern based on the received power satisfying a power threshold.
(63) In some examples of the method, apparatuses, and non-transitory computer-readable medium

described herein, the feedback information includes one or more negative acknowledgments received during a time period.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates an example of a system for wireless communications that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure.

(2) FIG. **2** illustrates an example of a wireless communications system that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure.

(3) FIGS. **3**A, **3**B, **3**C, **3**D, and **3**E illustrate examples of reference signal patterns that support reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure.

(4) FIGS. **4** through **7** illustrate examples of a process flow in a system that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure.

(5) FIGS. **8** and **9** show block diagrams of devices that support reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure.

(6) FIG. **10** shows a block diagram of a reference signal manager that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure.

(7) FIG. **11** shows a diagram of a system including a device that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure.

(8) FIGS. **12** through **18** show flowcharts illustrating methods that support reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

(9) Wireless communications devices operating in a system that supports sidelink communications (e.g., a vehicle-to-everything (V2X) system, a vehicle-to-vehicle (V2V) system, a cellular-V2X (C-V2X) system, or other type of device-to-device (D2D) system) may communicate with each other while in motion. For example, a UE within a V2X system may be traveling at a velocity and may attempt to transmit or receive communications with a neighboring UE (e.g., via sidelink transmission) traveling at a different velocity to exchange information. In some cases, the difference in the UE's velocities may span a wide range (e.g., a few kilometers per hour (kmph) to about 500 kmph), which may make control information transmissions inefficient (e.g., poor quality or excess overhead signaling). In some cases, efficient transmission of reference signals (e.g., demodulation reference signal (DMRS)) may vary based on the relative speed between the UEs. For instance, when a relative speed between the UEs is high, techniques may be used to ensure the reference signals are properly presented (e.g., a higher time-domain density of reference signals may be present). However, when the relative speed between the UEs is low, fewer reference signals in time-domain may need to be transmitted than at higher relative speeds. If the same number of reference signals are transmitted for high and low relative speeds, then either the reference signal pattern may be inefficient for high relative speeds or there may be unnecessary overhead for signaling under low relative speeds, or both. This may result in ineffective wireless spectral use and poor sidelink performance. Therefore, a flexible design of reference signal patterns to ensure

accurate and efficient decoding of control or data information (e.g., in channel estimation) at UEs of varying velocities may be beneficial.

(10) An adaptive design of DMRS time density based on UE velocity may allow for an increase in DMRS density when relative speed is high and a decrease in DMRS density when relative speed is low. The DMRS design may include a number of different DMRS patterns of varying density corresponding to different speed ranges (e.g., two, three, or four DMRS patterns and corresponding speed ranges). In some examples, a UE may send assistance information (e.g., UE's speed, direction, location, relative speed, subcarrier spacing, and/or DMRS pattern index) to the transmitting UE (which may be referred to herein as a "transmitter," a transmitting wireless device, or other like terminology) to help identify an efficient DMRS pattern based on the relative speed between the two UEs. The transmitter may then transmit the DMRS and data to a receiving UE (which may be referred to herein as a "receiver," a receiving wireless device, or other like terminology) after the transmitter determines the DMRS pattern. Assistance information may be transmitted periodically, after a change of assistance information, or based on a trigger from a transmitter.

(11) In another example, a UE may be unable to send or receive assistance information (e.g., UE's speed, direction, location, relative speed, subcarrier spacing, and/or DMRS pattern index). Thus, the transmitter may determine the DMRS pattern based on its own speed without information of the receiver's speed (e.g., in a broadcast transmission). The transmitter may then indicate the selected DMRS pattern in control information to one or more receivers followed by a transmission of the DMRS and data. In some cases, the UEs may utilize resources scheduled by a base station (e.g., when operating in mode 1 of sidelink communications). For instance, one or both of UEs may send assistance information (e.g., UE's speed, direction, location, relative speed, and/or DMRS pattern index) to the base station (e.g., gNB). The base station may determine the DMRS pattern to be used by the transmitting UE based on the received assistance information and then indicate the DMRS pattern to the transmitting UE. The transmitter may then transmit the DMRS, using the indicated pattern, and data.

(12) In another example, a transmitting UE may determine the DMRS pattern to use with the receiving UE based on hybrid automatic repeat request (HARQ) feedback from the receiving UE. The HARQ feedback may help determine a DMRS pattern by identifying an error (e.g., block error rate (BLER) or packet error rate) that may be compared to a threshold to determine the DMRS pattern for subsequent transmissions. For example, if the BLER is above a threshold, a denser DMRS pattern may be used for the next transmission in comparison to the previous transmission. Additionally or alternatively, a negative acknowledgement (NACK) may be used to determine received power at the transmitter, and the received power may be compared to a threshold to determine the DMRS pattern for subsequent transmissions. For instance, if the received power is above a threshold, a denser DMRS pattern may be used for the next transmission in comparison to the previous transmission. For example, in groupcast communications, multiple UEs may fail in decoding and send NACK to the transmitting UE. In such cases, the transmitting UE may receive a relatively larger NACK received signal power if more UEs send NACK at the same time (e.g., as compared to receiving a single NACK) because NACK signals may be combined over the air. That is, the received power of NACK at transmitter may be larger if more UEs send NACK. Accordingly, a large received power for NACK may indicate that the transmitting UE may use a denser DMRS pattern than the previous transmission. The transmitter may then transmit the DMRS and data after the transmitter determines the DMRS pattern. In some cases, the HARQ feedback may be stored over time such that the error rate or received power for a UE may be tracked over time.

(13) The UE may determine the DMRS pattern index using a mapping configuration. The mapping configuration may indicate a mapping between a DMRS pattern and a corresponding relative speed range or feedback thresholds. The mapping may be predefined (e.g., a table is stored at the UE),

preconfigured (multiple tables may be stored at the UE), or configured by higher layer signaling (e.g., radio resource control (RRC) signaling). Additionally or alternatively, the DMRS pattern may be determined based on a numerology (e.g., a subcarrier spacing) used within a system.

(14) Aspects of the disclosure are initially described in the context of wireless communications systems. Aspects of the disclosure are further illustrated by and described with reference to process flows, apparatus diagrams, system diagrams, and flowcharts that relate to transmitting DMRS for sidelink communications.

(15) FIG. **1** illustrates an example of a wireless communications system **100** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The wireless communications system **100** includes base stations **105**, UEs **115**, and a core network **130**. In some examples, the wireless communications system **100** may be a Long Term Evolution (LTE) network, an LTE-Advanced (LTE-A) network, an LTE-A Pro network, or a New Radio (NR) network. In some cases, wireless communications system **100** may support enhanced broadband communications, ultra-reliable (e.g., mission critical) communications, low latency communications, or communications with low-cost and low-complexity devices.

(16) Base stations **105** may wirelessly communicate with UEs **115** via one or more base station antennas. Base stations **105** described herein may include or may be referred to by those skilled in the art as a base transceiver station, a radio base station, an access point, a radio transceiver, a NodeB, an eNodeB (eNB), a next-generation NodeB or giga-NodeB (either of which may be referred to as a gNB), a Home NodeB, a Home eNodeB, or some other suitable terminology. Wireless communications system **100** may include base stations **105** of different types (e.g., macro or small cell base stations). The UEs **115** described herein may be able to communicate with various types of base stations **105** and network equipment including macro eNBs, small cell eNBs, gNBs, relay base stations, and the like.

(17) Each base station **105** may be associated with a particular geographic coverage area **110** in which communications with various UEs **115** is supported. Each base station **105** may provide communication coverage for a respective geographic coverage area **110** via communication links **125**, and communication links **125** between a base station **105** and a UE **115** may utilize one or more carriers. Communication links **125** shown in wireless communications system **100** may include uplink transmissions from a UE **115** to a base station **105**, or downlink transmissions from a base station **105** to a UE **115**. Downlink transmissions may also be called forward link transmissions while uplink transmissions may also be called reverse link transmissions.

(18) The geographic coverage area **110** for a base station **105** may be divided into sectors making up a portion of the geographic coverage area **110**, and each sector may be associated with a cell. For example, each base station **105** may provide communication coverage for a macro cell, a small cell, a hot spot, or other types of cells, or various combinations thereof. In some examples, a base station **105** may be movable and therefore provide communication coverage for a moving geographic coverage area **110**. In some examples, different geographic coverage areas **110** associated with different technologies may overlap and overlapping geographic coverage areas **110** associated with different technologies may be supported by the same base station **105** or by different base stations **105**. The wireless communications system **100** may include, for example, a heterogeneous LTE/LTE-A/LTE-A Pro or NR network in which different types of base stations **105** provide coverage for various geographic coverage areas **110**.

(19) The term “cell” refers to a logical communication entity used for communication with a base station **105** (e.g., over a carrier), and may be associated with an identifier for distinguishing neighboring cells (e.g., a physical cell identifier (PCID), a virtual cell identifier (VCID)) operating via the same or a different carrier. In some examples, a carrier may support multiple cells, and different cells may be configured according to different protocol types (e.g., machine-type communication (MTC), narrowband Internet-of-Things (NB-IoT), enhanced mobile broadband

(eMBB), or others) that may provide access for different types of devices. In some cases, the term "cell" may refer to a portion of a geographic coverage area **110** (e.g., a sector) over which the logical entity operates.

(20) UEs **115** may be dispersed throughout the wireless communications system **100**, and each UE **115** may be stationary or mobile. A UE **115** may also be referred to as a mobile device, a wireless device, a remote device, a handheld device, or a subscriber device, or some other suitable terminology, where the "device" may also be referred to as a unit, a station, a terminal, or a client. A UE **115** may also be a personal electronic device such as a cellular phone, a personal digital assistant (PDA), a tablet computer, a laptop computer, or a personal computer. In some examples, a UE **115** may also refer to a wireless local loop (WLL) station, an Internet of Things (IoT) device, an Internet of Everything (IoE) device, or an MTC device, or the like, which may be implemented in various articles such as appliances, vehicles, meters, or the like.

(21) Some UEs **115**, such as MTC or IoT devices, may be low cost or low complexity devices, and may provide for automated communication between machines (e.g., via Machine-to-Machine (M2M) communication). M2M communication or MTC may refer to data communication technologies that allow devices to communicate with one another or a base station **105** without human intervention. In some examples, M2M communication or MTC may include communications from devices that integrate sensors or meters to measure or capture information and relay that information to a central server or application program that can make use of the information or present the information to humans interacting with the program or application. Some UEs **115** may be designed to collect information or enable automated behavior of machines. Examples of applications for MTC devices include smart metering, inventory monitoring, water level monitoring, equipment monitoring, healthcare monitoring, wildlife monitoring, weather and geological event monitoring, fleet management and tracking, remote security sensing, physical access control, and transaction-based business charging.

(22) Some UEs **115** may be configured to employ operating modes that reduce power consumption, such as half-duplex communications (e.g., a mode that supports one-way communication via transmission or reception, but not transmission and reception simultaneously). In some examples, half-duplex communications may be performed at a reduced peak rate. Other power conservation techniques for UEs **115** include entering a power saving "deep sleep" mode when not engaging in active communications, or operating over a limited bandwidth (e.g., according to narrowband communications). In some cases, UEs **115** may be designed to support critical functions (e.g., mission critical functions), and a wireless communications system **100** may be configured to provide ultra-reliable communications for these functions.

(23) In some cases, a UE **115** may also be able to communicate directly with other UEs **115** (e.g., using a peer-to-peer (P2P) or device-to-device (D2D) protocol). One or more of a group of UEs **115** utilizing D2D communications may be within the geographic coverage area **110** of a base station **105**. Other UEs **115** in such a group may be outside the geographic coverage area **110** of a base station **105** or be otherwise unable to receive transmissions from a base station **105**. In some cases, groups of UEs **115** communicating via D2D communications may utilize a one-to-many (1:M) system in which each UE **115** transmits to every other UE **115** in the group. In some cases, a base station **105** facilitates the scheduling of resources for D2D communications. In other cases, D2D communications are carried out between UEs **115** without the involvement of a base station **105**.

(24) Base stations **105** may communicate with the core network **130** and with one another. For example, base stations **105** may interface with the core network **130** through backhaul links **132** (e.g., via an S1, N2, N3, or other interface). Base stations **105** may communicate with one another over backhaul links **134** (e.g., via an X2, Xn, or other interface) either directly (e.g., directly between base stations **105**) or indirectly (e.g., via core network **130**). A UE **115** may communicate with the core network **130** through communication link **135**.

(25) The core network **130** may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. The core network **130** may be an evolved packet core (EPC), which may include at least one mobility management entity (MME), at least one serving gateway (S-GW), and at least one Packet Data Network (PDN) gateway (P-GW). The MME may manage non-access stratum (e.g., control plane) functions such as mobility, authentication, and bearer management for UEs **115** served by base stations **105** associated with the EPC. User IP packets may be transferred through the S-GW, which itself may be connected to the P-GW. The P-GW may provide IP address allocation as well as other functions. The P-GW may be connected to the network operators IP services. The operators IP services may include access to the Internet, Intranet(s), an IP Multimedia Subsystem (IMS), or a Packet-Switched (PS) Streaming Service.
(26) At least some of the network devices, such as a base station **105**, may include subcomponents such as an access network entity, which may be an example of an access node controller (ANC). Each access network entity may communicate with UEs **115** through a number of other access network transmission entities, which may be referred to as a radio head, a smart radio head, or a transmission/reception point (TRP). In some configurations, various functions of each access network entity or base station **105** may be distributed across various network devices (e.g., radio heads and access network controllers) or consolidated into a single network device (e.g., a base station **105**).
(27) Wireless communications system **100** may operate using one or more frequency bands, typically in the range of 300 megahertz (MHz) to 300 gigahertz (GHz). Generally, the region from 300 MHz to 3 GHz is known as the ultra-high frequency (UHF) region or decimeter band, since the wavelengths range from approximately one decimeter to one meter in length. UHF waves may be blocked or redirected by buildings and environmental features. However, the waves may penetrate structures sufficiently for a macro cell to provide service to UEs **115** located indoors. Transmission of UHF waves may be associated with smaller antennas and shorter range (e.g., less than 100 km) compared to transmission using the smaller frequencies and longer waves of the high frequency (HF) or very high frequency (VHF) portion of the spectrum below 300 MHz.
(28) Wireless communications system **100** may also operate in a super high frequency (SHF) region using frequency bands from 3 GHz to 30 GHz, also known as the centimeter band. The SHF region includes bands such as the 5 GHz industrial, scientific, and medical (ISM) bands, which may be used opportunistically by devices that may be capable of tolerating interference from other users.
(29) Wireless communications system **100** may also operate in an extremely high frequency (EHF) region of the spectrum (e.g., from 30 GHz to 300 GHz), also known as the millimeter band. In some examples, wireless communications system **100** may support millimeter wave (mmW) communications between UEs **115** and base stations **105**, and EHF antennas of the respective devices may be even smaller and more closely spaced than UHF antennas. In some cases, this may facilitate use of antenna arrays within a UE **115**. However, the propagation of EHF transmissions may be subject to even greater atmospheric attenuation and shorter range than SHF or UHF transmissions. Techniques disclosed herein may be employed across transmissions that use one or more different frequency regions, and designated use of bands across these frequency regions may differ by country or regulating body.
(30) In some cases, wireless communications system **100** may utilize both licensed and unlicensed radio frequency spectrum bands. For example, wireless communications system **100** may employ License Assisted Access (LAA), LTE-Unlicensed (LTE-U) radio access technology, or NR technology in an unlicensed band such as the 5 GHz ISM band. When operating in unlicensed radio frequency spectrum bands, wireless devices such as base stations **105** and UEs **115** may employ listen-before-talk (LBT) procedures to ensure a frequency channel is clear before transmitting data. In some cases, operations in unlicensed bands may be based on a carrier aggregation configuration

in conjunction with component carriers operating in a licensed band (e.g., LAA). Operations in unlicensed spectrum may include downlink transmissions, uplink transmissions, peer-to-peer transmissions, or a combination of these. Duplexing in unlicensed spectrum may be based on frequency division duplexing (FDD), time division duplexing (TDD), or a combination of both.

(31) In some examples, base station **105** or UE **115** may be equipped with multiple antennas, which may be used to employ techniques such as transmit diversity, receive diversity, multiple-input multiple-output (MIMO) communications, or beamforming. For example, wireless communications system **100** may use a transmission scheme between a transmitting device (e.g., a base station **105**) and a receiving device (e.g., a UE **115**), where the transmitting device is equipped with multiple antennas and the receiving device is equipped with one or more antennas. MIMO communications may employ multipath signal propagation to increase the spectral efficiency by transmitting or receiving multiple signals via different spatial layers, which may be referred to as spatial multiplexing. The multiple signals may, for example, be transmitted by the transmitting device via different antennas or different combinations of antennas. Likewise, the multiple signals may be received by the receiving device via different antennas or different combinations of antennas. Each of the multiple signals may be referred to as a separate spatial stream and may carry bits associated with the same data stream (e.g., the same codeword) or different data streams. Different spatial layers may be associated with different antenna ports used for channel measurement and reporting. MIMO techniques include single-user MIMO (SU-MIMO) where multiple spatial layers are transmitted to the same receiving device, and multiple-user MIMO (MU-MIMO) where multiple spatial layers are transmitted to multiple devices.

(32) Beamforming, which may also be referred to as spatial filtering, directional transmission, or directional reception, is a signal processing technique that may be used at a transmitting device or a receiving device (e.g., a base station **105** or a UE **115**) to shape or steer an antenna beam (e.g., a transmit beam or receive beam) along a spatial path between the transmitting device and the receiving device. Beamforming may be achieved by combining the signals communicated via antenna elements of an antenna array such that signals propagating at particular orientations with respect to an antenna array experience constructive interference while others experience destructive interference. The adjustment of signals communicated via the antenna elements may include a transmitting device or a receiving device applying certain amplitude and phase offsets to signals carried via each of the antenna elements associated with the device. The adjustments associated with each of the antenna elements may be defined by a beamforming weight set associated with a particular orientation (e.g., with respect to the antenna array of the transmitting device or receiving device, or with respect to some other orientation).

(33) In one example, a base station **105** may use multiple antennas or antenna arrays to conduct beamforming operations for directional communications with a UE **115**. For instance, some signals (e.g., synchronization signals, reference signals, beam selection signals, or other control signals) may be transmitted by a base station **105** multiple times in different directions, which may include a signal being transmitted according to different beamforming weight sets associated with different directions of transmission. Transmissions in different beam directions may be used to identify (e.g., by the base station **105** or a receiving device, such as a UE **115**) a beam direction for subsequent transmission and/or reception by the base station **105**.

(34) Some signals, such as data signals associated with a particular receiving device, may be transmitted by a base station **105** in a single beam direction (e.g., a direction associated with the receiving device, such as a UE **115**). In some examples, the beam direction associated with transmissions along a single beam direction may be determined based at least in in part on a signal that was transmitted in different beam directions. For example, a UE **115** may receive one or more of the signals transmitted by the base station **105** in different directions, and the UE **115** may report to the base station **105** an indication of the signal it received with a highest signal quality, or an otherwise acceptable signal quality. Although these techniques are described with reference to

signals transmitted in one or more directions by a base station **105**, a UE **115** may employ similar techniques for transmitting signals multiple times in different directions (e.g., for identifying a beam direction for subsequent transmission or reception by the UE **115**), or transmitting a signal in a single direction (e.g., for transmitting data to a receiving device).

(35) A receiving device (e.g., a UE **115**, which may be an example of a mmW receiving device) may try multiple receive beams when receiving various signals from the base station **105**, such as synchronization signals, reference signals, beam selection signals, or other control signals. For example, a receiving device may try multiple receive directions by receiving via different antenna subarrays, by processing received signals according to different antenna subarrays, by receiving according to different receive beamforming weight sets applied to signals received at a plurality of antenna elements of an antenna array, or by processing received signals according to different receive beamforming weight sets applied to signals received at a plurality of antenna elements of an antenna array, any of which may be referred to as “listening” according to different receive beams or receive directions. In some examples, a receiving device may use a single receive beam to receive along a single beam direction (e.g., when receiving a data signal). The single receive beam may be aligned in a beam direction determined based listening according to different receive beam directions (e.g., a beam direction determined to have a highest signal strength, highest signal-to-noise ratio, or otherwise acceptable signal quality based listening according to multiple beam directions).

(36) In some cases, the antennas of a base station **105** or UE **115** may be located within one or more antenna arrays, which may support MIMO operations, or transmit or receive beamforming. For example, one or more base station antennas or antenna arrays may be co-located at an antenna assembly, such as an antenna tower. In some cases, antennas or antenna arrays associated with a base station **105** may be located in diverse geographic locations. A base station **105** may have an antenna array with a number of rows and columns of antenna ports that the base station **105** may use to support beamforming of communications with a UE **115**. Likewise, a UE **115** may have one or more antenna arrays that may support various MIMO or beamforming operations.

(37) In some cases, wireless communications system **100** may be a packet-based network that operate according to a layered protocol stack. In the user plane, communications at the bearer or Packet Data Convergence Protocol (PDCP) layer may be IP-based. A Radio Link Control (RLC) layer may perform packet segmentation and reassembly to communicate over logical channels. A Medium Access Control (MAC) layer may perform priority handling and multiplexing of logical channels into transport channels. The MAC layer may also use HARQ to provide retransmission at the MAC layer to improve link efficiency. In the control plane, the Radio Resource Control (RRC) protocol layer may provide establishment, configuration, and maintenance of an RRC connection between a UE **115** and a base station **105** or core network **130** supporting radio bearers for user plane data. At the Physical layer, transport channels may be mapped to physical channels.

(38) In some cases, UEs **115** and base stations **105** may support retransmissions of data to increase the likelihood that data is received successfully. HARQ feedback is one technique of increasing the likelihood that data is received correctly over a communication link **125**. HARQ may include a combination of error detection (e.g., using a cyclic redundancy check (CRC)), forward error correction (FEC), and retransmission (e.g., automatic repeat request (ARQ)). HARQ may improve throughput at the MAC layer in poor radio conditions (e.g., signal-to-noise conditions). In some cases, a wireless device may support same-slot HARQ feedback, where the device may provide HARQ feedback in a specific slot for data received in a previous symbol in the slot. In other cases, the device may provide HARQ feedback in a subsequent slot, or according to some other time interval.

(39) Time intervals in LTE or NR may be expressed in multiples of a basic time unit, which may, for example, refer to a sampling period of T.sub.s=1/30,720,000 seconds. Time intervals of a communications resource may be organized according to radio frames each having a duration of 10

milliseconds (ms), where the frame period may be expressed as T.sub.f=307,200 T.sub.s. The radio frames may be identified by a system frame number (SFN) ranging from 0 to 1023. Each frame may include 10 subframes numbered from 0 to 9, and each subframe may have a duration of 1 ms. A subframe may be further divided into 2 slots each having a duration of 0.5 ms, and each slot may contain 6 or 7 modulation symbol periods (e.g., depending on the length of the cyclic prefix prepended to each symbol period). Excluding the cyclic prefix, each symbol period may contain 2048 sampling periods. In some cases, a subframe may be the smallest scheduling unit of the wireless communications system **100** and may be referred to as a transmission time interval (TTI). In other cases, a smallest scheduling unit of the wireless communications system **100** may be shorter than a subframe or may be dynamically selected (e.g., in bursts of shortened TTIs (sTTIs) or in selected component carriers using sTTIs).
(40) In some wireless communications systems, a slot may further be divided into multiple mini-slots containing one or more symbols. In some instances, a symbol of a mini-slot or a mini-slot may be the smallest unit of scheduling. Each symbol may vary in duration depending on the subcarrier spacing or frequency band of operation, for example. Further, some wireless communications systems may implement slot aggregation in which multiple slots or mini-slots are aggregated together and used for communication between a UE **115** and a base station **105**.
(41) The term “carrier” refers to a set of radio frequency spectrum resources having a defined physical layer structure for supporting communications over a communication link **125**. For example, a carrier of a communication link **125** may include a portion of a radio frequency spectrum band that is operated according to physical layer channels for a given radio access technology. Each physical layer channel may carry user data, control information, or other signaling. A carrier may be associated with a pre-defined frequency channel (e.g., an evolved universal mobile telecommunication system terrestrial radio access (E-UTRA) absolute radio frequency channel number (EARFCN)) and may be positioned according to a channel raster for discovery by UEs **115**. Carriers may be downlink or uplink (e.g., in an FDD mode), or be configured to carry downlink and uplink communications (e.g., in a TDD mode). In some examples, signal waveforms transmitted over a carrier may be made up of multiple sub-carriers (e.g., using multi-carrier modulation (MCM) techniques such as orthogonal frequency division multiplexing (OFDM) or discrete Fourier transform spread OFDM (DFT-S-OFDM)).
(42) The organizational structure of the carriers may be different for different radio access technologies (e.g., LTE, LTE-A, LTE-A Pro, NR). For example, communications over a carrier may be organized according to TTIs or slots, each of which may include user data as well as control information or signaling to support decoding the user data. A carrier may also include dedicated acquisition signaling (e.g., synchronization signals or system information, etc.) and control signaling that coordinates operation for the carrier. In some examples (e.g., in a carrier aggregation configuration), a carrier may also have acquisition signaling or control signaling that coordinates operations for other carriers.
(43) Physical channels may be multiplexed on a carrier according to various techniques. A physical control channel and a physical data channel may be multiplexed on a downlink carrier, for example, using time division multiplexing (TDM) techniques, frequency division multiplexing (FDM) techniques, or hybrid TDM-FDM techniques. In some examples, control information transmitted in a physical control channel may be distributed between different control regions in a cascaded manner (e.g., between a common control region or common search space and one or more UE-specific control regions or UE-specific search spaces).
(44) A carrier may be associated with a particular bandwidth of the radio frequency spectrum, and in some examples, the carrier bandwidth may be referred to as a “system bandwidth” of the carrier or the wireless communications system **100**. For example, the carrier bandwidth may be one of a number of predetermined bandwidths for carriers of a particular radio access technology (e.g., 1.4, 3, 5, 10, 15, 20, 40, or 80 MHz). In some examples, each served UE **115** may be configured for

operating over portions or all of the carrier bandwidth. In other examples, some UEs **115** may be configured for operation using a narrowband protocol type that is associated with a predefined portion or range (e.g., set of subcarriers or RBs) within a carrier (e.g., "in-band" deployment of a narrowband protocol type).
(45) In a system employing MCM techniques, a resource element may consist of one symbol period (e.g., a duration of one modulation symbol) and one subcarrier, where the symbol period and subcarrier spacing are inversely related. The number of bits carried by each resource element may depend on the modulation scheme (e.g., the order of the modulation scheme). Thus, the more resource elements that a UE **115** receives and the higher the order of the modulation scheme, the higher the data rate may be for the UE **115**. In MIMO systems, a wireless communications resource may refer to a combination of a radio frequency spectrum resource, a time resource, and a spatial resource (e.g., spatial layers), and the use of multiple spatial layers may further increase the data rate for communications with a UE **115**.
(46) Devices of the wireless communications system **100** (e.g., base stations **105** or UEs **115**) may have a hardware configuration that supports communications over a particular carrier bandwidth or may be configurable to support communications over one of a set of carrier bandwidths. In some examples, the wireless communications system **100** may include base stations **105** and/or UEs **115** that support simultaneous communications via carriers associated with more than one different carrier bandwidth.
(47) Wireless communications system **100** may support communication with a UE **115** on multiple cells or carriers, a feature which may be referred to as carrier aggregation or multi-carrier operation. A UE **115** may be configured with multiple downlink component carriers and one or more uplink component carriers according to a carrier aggregation configuration. Carrier aggregation may be used with both FDD and TDD component carriers.
(48) In some cases, wireless communications system **100** may utilize enhanced component carriers (eCCs). An eCC may be characterized by one or more features including wider carrier or frequency channel bandwidth, shorter symbol duration, shorter TTI duration, or modified control channel configuration. In some cases, an eCC may be associated with a carrier aggregation configuration or a dual connectivity configuration (e.g., when multiple serving cells have a suboptimal or non-ideal backhaul link). An eCC may also be configured for use in unlicensed spectrum or shared spectrum (e.g., where more than one operator is allowed to use the spectrum). An eCC characterized by wide carrier bandwidth may include one or more segments that may be utilized by UEs **115** that are not capable of monitoring the whole carrier bandwidth or are otherwise configured to use a limited carrier bandwidth (e.g., to conserve power).
(49) In some cases, an eCC may utilize a different symbol duration than other component carriers, which may include use of a reduced symbol duration as compared with symbol durations of the other component carriers. A shorter symbol duration may be associated with increased spacing between adjacent subcarriers. A device, such as a UE **115** or base station **105**, utilizing eCCs may transmit wideband signals (e.g., according to frequency channel or carrier bandwidths of 20, 40, 60, 80 MHz, etc.) at reduced symbol durations (e.g., 16.67 microseconds). A TTI in eCC may consist of one or multiple symbol periods. In some cases, the TTI duration (that is, the number of symbol periods in a TTI) may be variable.
(50) Wireless communications system **100** may be an NR system that may utilize any combination of licensed, shared, and unlicensed spectrum bands, among others. The flexibility of eCC symbol duration and subcarrier spacing may allow for the use of eCC across multiple spectrums. In some examples, NR shared spectrum may increase spectrum utilization and spectral efficiency, specifically through dynamic vertical (e.g., across the frequency domain) and horizontal (e.g., across the time domain) sharing of resources.
(51) In a V2X communications system, sidelink communications **140** (e.g., control or data transmissions) between UEs **115** may occur while one or both UEs **115** are moving. For example,

UE **115** within a V2X system may be traveling at a velocity and may attempt to transmit or receive communications with a neighboring UE **115** (e.g., via sidelink transmission) traveling at a different velocity. In some cases, the difference in the UE's **115** velocities may span a wide range (e.g., a few kmph to about 500 kmph). This wide range of relative speed between the UEs **115** may make control information transmissions inefficient because efficient transmission of reference signals (e.g., DMRS) may vary based on the relative speed between the UEs **115**. For instance, when relative speed between the UEs **115** is high, more reference signals in time may be utilized to ensure data is properly received (e.g., a higher density of reference signals may be present). However, when relative speed between the UEs **115** is low, fewer reference signals may need to be transmitted than at higher relative speeds. As such, a UE **115** may employ enhanced DMRS design to ensure efficient spectral use and improve sidelink performance.
(52) An adaptive design of DMRS time density based on UE **115** velocity may allow for an increase in DMRS density in sidelink communications **140** when relative speed is high and a decrease in DMRS density when relative speed is low. The DMRS design may include a number of different DMRS patterns of varying density corresponding to different speed ranges (e.g., two, three, or four DMRS patterns and corresponding speed ranges), which will be discussed below in greater detail. In some examples, a UE **115** may send assistance information (e.g., UE's **115** speed, direction, location, relative speed, subcarrier spacing, and/or DMRS pattern index) to the transmitting UE **115** to help identify an efficient DMRS pattern based on the relative speed between the two UEs **115**. The transmitter may then transmit the DMRS and data via sidelink communications **140** after the transmitter determines the DMRS pattern. Assistance information may be transmitted periodically, after a change of assistance information, or based on a trigger from a transmitter.
(53) In another example, a UE **115** may be unable to send or receive assistance information (e.g., UE's **115** speed, direction, location, relative speed, subcarrier spacing, and/or DMRS pattern index). Thus, the transmitting UE **115** may determine the DMRS pattern based on its speed and/or the subcarrier spacing without information of the receiver's speed (e.g., in a broadcast system). The transmitter may then indicate the selected DMRS pattern in control information via sidelink communications **140** to the receiver followed by a transmission of the DMRS and data sidelink communications **140**. In some cases, the UEs **115** may follow a sidelink schedule for sidelink communications **140** determined by a base station **105** (e.g., when UEs **115** are operating in mode 1). For instance, one or both of UEs **115** may send assistance information (e.g., UE's **115** speed, direction, location, relative speed, and/or DMRS pattern index) to the base station **105** (e.g., gNB). The base station **105** may determine the DMRS pattern to be used by the transmitting UE **115** based on the received assistance information and then indicate the DMRS pattern to the transmitting UE **115** via link **125**. The transmitting UE **115** may then transmit the DMRS, using the indicated pattern, and data using sidelink communications **140**.
(54) In another example, a transmitting UE **115** may determine the DMRS pattern to use with the receiving UE **115** based on HARQ feedback from the receiving UE **115**. The HARQ feedback may help determine a DMRS pattern by identifying an error (e.g., BLER or packed error) that may be compared to a threshold to determine the DMRS pattern for subsequent transmissions on sidelink communications **140**. For example, if the BLER is above a threshold, a denser DMRS pattern may be used for the next transmission in comparison to the previous transmission. Additionally or alternatively, a NACK may be used to determine received power at the transmitting UE **115**, and the received power may be compared to a threshold to determine the DMRS pattern for subsequent transmissions on sidelink communications **140**. For instance, if the received power is above a threshold, a denser DMRS pattern may be used for the next transmission in comparison to the previous transmission. The transmitting UE **115** may then transmit the DMRS and data via sidelink communications **140** after the transmitting UE **115** determines the DMRS pattern. In some cases, the HARQ feedback may be stored over time such that the error rate or received power for UE **115**

may be tracked over time.

(55) UE **115** may determine the DMRS pattern index using a mapping configuration. The mapping configuration may indicate a mapping between a DMRS pattern and a corresponding relative speed range or feedback thresholds. The mapping may be predefined (e.g., a table is stored at UE **115**), preconfigured (e.g., multiple tables may be stored at UE **115** where a mapping in the table may be preconfigured, or multiple tables may be specified, where one of the tables may be preconfigured for use), or configured by higher layer signaling (e.g., RRC signaling).

(56) UEs **115** may include a reference signal manager **101**, which may enable a UE **115** to identify a mapping configuration to use for determining DMRS patterns for relative speed ranges. The UE **115** may transmit or receive assistance information to determine the DMRS pattern for current operating conditions of UE **115**. In some examples, the UE **115** may transmit data including the DMRS pattern, for example, in a sidelink communication **140** to another UE **115**. In some examples, the reference signal manager **101** may transmit assistance information to a second wireless device, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof and identify a reference signal pattern of a reference signal for demodulating data, the reference signal pattern based on the assistance information. In some cases, the reference signal manager **101** may also determine a speed of the first wireless device, identify a subcarrier spacing for communicating with one or more other wireless devices, determine, based on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data, and transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.

(57) Additionally or alternatively, the reference signal manager **101** may also transmit assistance information to a base station, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof, receive, from the base station, an indication of a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the assistance information, and transmit the data and the reference signal to a second wireless device, the reference signal being transmitted in accordance with the reference signal pattern. The reference signal manager **101** may receive, from one or more other wireless devices, feedback information in response to one or more previous data transmissions to the one or more other wireless devices, determine a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the feedback information, and transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.

(58) FIG. **2** illustrates an example of a wireless communications system **200** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. In some examples, wireless communications system **200** may implement aspects of wireless communications system **100**. In some examples, the wireless communications system **200** may include UEs **115**-*a* and **115**-*b*, and base station **105**-*a* which may be examples of a UE **115** and base station **105**, respectively, described with reference to FIG. **1**. It is noted that communications between two UEs **115** are illustrated in wireless communications system **200** for the sake of brevity, and the techniques described herein may be applicable to multiple UEs **115** within a system. For instance, UE **115**-*a* may communicate with multiple UEs **115**, for example, using broadcast or groupcast communications schemes.

(59) In some cases, the UEs **115**-*a* and **115**-*b* may communicate with each other within a V2X system (e.g., using the sidelink communications **205**) and may employ a flexible DMRS pattern design to enable efficient sidelink communications. For example, efficient transmissions of reference signals (e.g., DMRS) may vary based on the relative speed between UEs **115**-*a* and **115**-*b*. As described herein, the adaptive time density of DMRS based on UE **115**-*a* or **115**-*b* velocity may allow for an increase in DMRS density when relative speed between UEs **115**-*a* and **115**-*b* is

high and a decrease in DMRS density when relative speed between UEs **115**-*a* and **115**-*b* is low. Further, system **200** may be divided into geographical zones (e.g., zone **215** and **220**), shown by the dotted lines.

(60) According to some aspects, the DMRS design may include a number of different DMRS patterns of varying density corresponding to different speed categories (e.g., two, three, or four DMRS patterns and corresponding speed ranges) to be used for data channel estimation. For instance, UE **115**-*a* may attempt to decode sidelink communications **205** (e.g., data transmissions) from the UE **115**-*b* when one or both of UEs **115**-*a* and **115**-*b* are in motion. For example, UE **115**-*b* within wireless communications system **200** may be traveling at a relatively low speed (e.g., 10 kmph in a direction) to point **210** and may attempt to transmit or receive communications with a stationary neighboring UE **115**-*a* (e.g., via sidelink transmission **205**). Assistance information (e.g., a speed, direction, location (e.g., zone ID), relative speed, subcarrier spacing, and/or DMRS pattern index of a UE **115**) from one or both UEs **115**-*a* and **115**-*b* may be used to determine the relative speed between UEs **115**-*a* and **115**-*b*. In some cases, the difference in the velocities of the UEs **115** (e.g., relative speed between one another) may be categorized (e.g., mapped) into a speed category (e.g., low: 0-60 kmph, medium: 61-120 kmph, or high: greater than 120 kmph), where a different DMRS pattern may correspond to each speed category (e.g., low, medium, and high). Example DMRS patterns are shown in FIG. **3** and described in greater detail herein. The variable DMRS design (e.g., use of assistance information) may vary based on the system's **200** configurations (e.g., unicast, groupcast, or broadcast).

(61) In some examples, UE **115**-*a* may send assistance information (e.g., UE's **115**-*a* speed, direction, location (e.g., zone ID), relative speed, subcarrier spacing, and/or DMRS pattern index) to the transmitting UE **115**-*b* to help identify an efficient DMRS pattern based on the relative speed between the two UEs **115**-*a* and **115**-*b*. In some examples, assistance information may be transmitted in sidelink channel information (SCI), a sidelink data channel (SL-SCH) (e.g., MAC control element (CE)), or a feedback channel. In some cases, relative speed and/or a DMRS pattern index may be determined at UE **115**-*a* when UE **115**-*a* has assistance information about UE **115**-*b*. The transmitting UE **115**-*b* may then transmit the DMRS and data after the transmitting UE **115**-*b* determines the DMRS pattern based on the assistance information. Assistance information may be transmitted periodically (e.g., according to a schedule), after a change of assistance information (e.g., change in speed and/or direction of UE **115**-*a* or **115**-*b*), or based on a trigger from a transmitting UE **115**-*b*. This example is described in greater detail in FIG. **4**.

(62) In another example, UE **115**-*b* may operate in a broadcast configuration and may not receive transmissions from the receiver UE **115**-*a*. Accordingly, UE **115**-*b* may be unable to receive assistance information (e.g., UE's speed, direction, location, relative speed, subcarrier spacing, and/or DMRS pattern index) from UE **115**-*a*. Thus, the transmitting UE **115**-*b* may determine the DMRS pattern based on its speed without information of the receiving UE's **115**-*a* speed. The transmitting UE **115**-*b* may then indicate the selected DMRS pattern in control information via sidelink **205** to the receiving UE **115**-*a* followed by a transmission of the DMRS and data via sidelink **205**. This example is described in greater detail in FIG. **5**.

(63) In some cases, the UEs **115**-*a* and **115**-*b* may utilize resources (e.g., over sidelink **205**) scheduled by base station **105**-*a* (e.g., when operating in mode 1 of a sidelink communications scheme). In such cases, one or both of UEs **115**-*a* and **115**-*b* may send assistance information (e.g., UE's own speed, direction, location, relative speed, and/or DMRS pattern index) to the base station **105**-*a* (e.g., gNB) via links **225**. In some examples, assistance information may be transmitted in uplink channel information (UCI), a data channel (PUSCH) (e.g., MAC control element (CE)), or a feedback channel. The base station **105**-*a* may then determine the DMRS pattern to be used by the transmitting UE **115**-*a* based on the received assistance information and then indicate the DMRS pattern to the transmitting UE **115**-*a* via link **225**. The transmitting UE **115**-*a* may then transmit the DMRS, using the indicated pattern, and data to the receiving UE **115**-*b* via sidelink **205**. This

example is described in greater detail in FIG. **6**.
(64) In another example, a transmitting UE **115**-*a* may determine the DMRS pattern to use with the receiving UE **115**-*b* based on HARQ feedback from the receiving UE **115**-*b*. The HARQ feedback may help determine a DMRS pattern by identifying an error (e.g., BLER or packed error) that may be compared to a threshold to determine the DMRS pattern for subsequent transmissions via sidelink **205** (e.g., in a unicast configuration). For example, if the BLER satisfies a threshold, a denser DMRS pattern may be used for the next transmission via sidelink **205** in comparison to the previous transmission via sidelink **205**. Additionally or alternatively, a negative acknowledgement (NACK) may be used to determine received power at the transmitting UE **115**-*a*, and the received power may be compared to a threshold to determine the DMRS pattern for subsequent transmissions via sidelink **205** (e.g., in a groupcast configuration). For instance, if the received power is above a threshold, a denser DMRS pattern may be used for the next transmission via sidelink **205** in comparison to the previous transmission via sidelink **205**. For example, in a groupcast deployment, multiple UEs **115** may fail in decoding and send feedback information (e.g., NACK) to the transmitting UE **115**-*a*. In cases, where the transmitting UE **115**-*a* receives a larger NACK received signal power (e.g., because NACK signals may be combined over the air), the transmitting UE **115**-*a* may determine that a denser DMRS pattern may be used (e.g., as compared to the DMRS pattern used for the previous transmission). The transmitting UE **115**-*a* may then transmit the DMRS and data via sidelink **205** after the transmitting UE **115**-*a* determines the DMRS pattern. In some cases, the HARQ feedback may be stored over time such that the error rate or received power for a UE may be tracked over time. This example is described in greater detail in FIG. **7**.
(65) The UEs **115**-*a* and **115**-*b*, and base station **105**-*a* may determine the DMRS pattern index using a mapping configuration. The mapping configuration may indicate a one-to-one mapping between a DMRS pattern and a corresponding relative speed range or feedback thresholds. For example, two speed categories and DMRS patterns may be configured, such as a low speed category (e.g., 0 to 120 kmph) may correspond to a low density DMRS pattern, while a high speed category (e.g., greater than 120 kmph) may correspond to a high density DMRS pattern. The mapping may be predefined (e.g., a table is stored at the UEs **115**-*a* and **115**-*b*), preconfigured (multiple tables may be stored at the UEs **115**-*a* and **115**-*b*), or configured by higher layer signaling (e.g., RRC signaling from base station **105**-*a*).
(66) In some examples, a DMRS pattern may be determined for more than one UE **115** when the system is operating in a broadcast or groupcast configuration. Each receiving UE **115** may be moving at different speed, but one DMRS pattern may be used for all UEs **115**. The DMRS pattern may be based on assistance information from one or more UEs **115** and may account for the different conditions of each sidelink **205** with each UE **115**. For instance, a DMRS pattern may be selected based on the worst sidelink **205** (e.g., highest relative speed between UEs **115**). For example, the DMRS pattern with the highest DMRS time density of indicated DMRS patterns from receiving UEs **115** may be selected.
(67) Alternatively, a DMRS pattern may be selected based on the worst sidelink **205** and a BLER target. For example, the DMRS pattern may be selected within an error tolerance (e.g., 5% or 15%) such that not every receiving UE **115** is ensured successful decoding of the data channel. For example, transmitting UE **115** receives assistance info from 30 receiving UEs **115** and determines the relative speed for the 30 links based on three speed categories: low, medium, and high. 28 of the links within the low to medium relative speed categories, and 2 within the high relative speed category. If the BLER target is 0.1 (i.e., tolerance to 10% error), then the DMRS pattern may be selected based on the medium speed category. Therefore, the two UEs **115** within the high speed category may not decode correctly, but the BLER target can still be met.
(68) The DMRS pattern may be based on subcarrier spacing since a larger subcarrier spacing may correspond to a smaller (e.g., shorter) OFDM symbol duration, and a DMRS pattern that works for

larger subcarrier spacing may not work for smaller subcarrier spacing because a smaller subcarrier spacing may need a denser DMRS pattern than larger subcarrier spacing, even within the same or similar relative speeds. For example, when the transmitting UE **115** is transmitting with 30 kHz subcarrier spacing, a pattern with DMRS in three symbols (e.g., as described herein and shown in FIG. **3**D) may be determined (medium speed category) for a relative speed of 100 kmph. However, when the transmitting UE **115** is transmitting with 15 kHz subcarrier spacing, a reference signal pattern with DMRS in six symbols (e.g., as described herein and as shown in FIG. **3**E) may be determined (high speed category) for a relative speed of 100 kmph (e.g., due to a relatively longer duration of a slot). In another example, when the transmitting UE **115** is transmitting with 60 kHz subcarrier spacing, the pattern with DMRS in two symbols (e.g., as described herein and as shown in FIG. **3**C) may be determined (low speed category) for a relative speed of 100 kmph (e.g., due to a relatively shorter duration of a slot).

(69) FIGS. **3**A through **3**E illustrate an example of reference signal patterns **300** that support reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. In some examples, the DMRS patterns **300**-*a* through **300**-*e* may implement aspects of wireless communications systems **100** or **200** or may be implemented by a UE **115** such as a vehicle (or other wireless device in a V2X system), as described herein.

(70) As shown in FIG. **3**A, the reference signal pattern **300**-*a* (e.g., a DMRS pattern) may include OFDM symbols **305** with DMRS **310** or without DMRS **310**. For example, symbol **315** may be an example of an OFDM symbol **305** without DMRS **310**, and symbol **320** may be an example of an OFDM symbol **305** with DMRS **310**. In some examples, a symbol **320** with DMRS **310** may be transmitted every seventh symbol, such that there are six symbols **315** without DMRS **310** between each symbol **320** with DMRS **310**. In some cases, more symbols **315** without DMRS **310** may be present between each symbol **320** with DMRS **310** (e.g., as compared to other DMRS patterns **300**).

(71) As shown in FIG. **3**B, the reference signal pattern **300**-*b* may include OFDM symbols **305** with DMRS **310** or without DMRS **310**. For example, symbol **315** may be an example of a symbol **305** without DMRS **310**, and symbol **320** may be an example of a symbol **305** with DMRS **310**. In some examples, a symbol **320** with DMRS **310** may be transmitted every third or fourth symbol such that there are two or three symbols **315** without DMRS **310** between each symbol **320** with DMRS **310**, respectively.

(72) In some examples, the reference signal patterns **300**-*a* and **300**-*b* may be used when two speed categories are used. For instance, a low speed category (e.g., 0 to 100 kmph) may correspond to DMRS pattern **300**-*a*, while a high speed category (e.g., greater than 100 kmph) may correspond to DMRS pattern **300**-*b*. This example does not limit the number of categories to two, and more or less categories and corresponding DMRS patterns may be used. For example, three categories are described with reference to FIGS. **3**C-**3**E. In some examples, the mapping described here corresponds to a specific subcarrier spacing, for example, for 30 kHz subcarrier spacing. When subcarrier spacing is different is different, the mapping is different. For example, when subcarrier spacing is 60 kHz, pattern **300**-*b* corresponds to a high speed category (e.g., greater than 200 kmph).

(73) As shown in FIG. **3**C, the reference signal pattern **300**-*c* may include OFDM symbols **305** with DMRS **310** or without DMRS **310**. For example, symbol **315** may be an example of a symbol **305** without DMRS **310**, and symbol **320** may be an example of a symbol **305** with DMRS **310**. In some examples, a symbol **320** with DMRS **310** may be transmitted every tenth symbol such that there are nine symbols **315** without DMRS **310** between each symbol **320** with DMRS **310**, respectively.

(74) As shown in FIG. **3**D, the reference signal pattern **300**-*d* may include OFDM symbols **305** with DMRS **310** or without DMRS **310**. For example, symbol **315** may be an example of a symbol

305 without DMRS **310**, and symbol **320** may be an example of a symbol **305** with DMRS **310**. In some examples, a symbol **320** with DMRS **310** may be transmitted every fifth symbol such that there are four symbols **315** without DMRS **310** between each symbol **320** with DMRS **310**, respectively.

(75) As shown in FIG. **3**E, the reference signal pattern **300**-*e* may include OFDM symbols **305** with DMRS **310** or without DMRS **310**. For example, symbol **315** may be an example of a symbol **305** without DMRS **310**, and symbol **320** may be an example of a symbol **305** with DMRS **310**. In some examples, a symbol **320** with DMRS **310** may be transmitted every other symbol such that there is one symbol **315** without DMRS **310** between each symbol **320** with DMRS **310**, respectively. The DMRS pattern may last for one or more slots (e.g., fourteen symbols).

(76) In some examples, the reference signal patterns **300**-*c*, **300**-*d*, and **300**-*e* may be used when three speed categories are used. For instance, a low speed category (e.g., 0 to 60 kmph) may correspond to DMRS pattern **300**-*c*, while a medium speed category (e.g., 60 to 150 kmph) may correspond to DMRS pattern **300**-*d*, and a high speed category (e.g., greater than 150 kmph) may correspond to DMRS pattern **300**-*e*. This example does not limit the number of categories to three, and more or less categories and corresponding DMRS patterns may be used. For example, four speed categories may be used corresponding to DMRS patterns **300**-*c*, **300**-*a*, **300**-*d*, and **300**-*e*, from lowest to highest speed categories respectively. Additional categories or mapping may be used for different subcarrier spacing. For example, mapping may be based on speed and/or subcarrier spacing, such that mapping may be different from categories containing only speed. For instance, the range of a speed category may be different when subcarrier spacing is considered or each subcarrier spacing may have a mapping table as an alternative or in addition to the speed mapping tables.

(77) FIG. **4** illustrates an example of a process flow **400** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. In some examples, process flow **400** may implement aspects of wireless communications system **100**. Process flow **400** may be implemented by a UE **115**-*c* and UE **115**-*d*, or any other examples of UEs **115** as described herein. Alternative examples of the following may be implemented, where some steps are performed in a different order than described or are not performed at all. In some cases, steps may include additional features not mentioned below, or further steps may be added.

(78) At **405**, UE **115**-*c* may optionally receive transmitter assistance (e.g., UE's **115**-*d* speed, direction, location (e.g., zone ID), relative speed, subcarrier spacing, and/or DMRS pattern index) information from UE **115**-*d*. The transmitter assistance information may be transmitted for other purposes than DMRS pattern design determination but may still be used by UE **115**-*c*, which may reduce signal overhead between UE **115**-*c* and UE **115**-*d* by reducing the addition of DMRS pattern design specific assistance information.

(79) In some examples, assistance information may be transmitted in SCI, a sidelink data channel (SL-SCH) (e.g., MAC CE), or a feedback channel. Assistance information may be transmitted periodically (e.g., according to a predefined or preconfigured time period, indicated by RRC signaling), after a change of assistance information (e.g., change in speed, direction, location and/or relative speed of UE **115**-*c* or **115**-*d*), or based on a trigger from a receiving UE **115**-*c*. More specifically, a change of assistance information at UE **115**-*d* may include when the change of speed exceeds a threshold (e.g., a change of absolute value is larger than a threshold, or change from one speed category to another: low to medium), a change of direction exceeds a threshold, the location is changed (e.g., zone ID changed), the change of relative speed exceeds a threshold (if UE **115**-*d* determines relative speed of UEs **115**-*c* and **115**-*d*), and/or the determined DMRS pattern is changed (if UE **115**-*d* determines DMRS pattern).

(80) In some cases, speed may be indicated by an index mapped from absolute speed. For example, three indices to indicate low, medium, and high speed, and each of the three indices corresponds to

a speed range (e.g., low: less than or equal to 60 kmph, mid: greater than 60 and less than or equal to 120 kmph, and high: greater than 120 kmph). In some examples, direction may be indicated by an index mapped from UE's **115** absolute direction. For instance, the degree from north direction, rounded to a finite set of directions (e.g., eight directions mapped from [0, 360) degrees, resulting in eight equal ranges of 45 degrees).

(81) In some examples, speed and/or direction may be determined by a UE's **115** location (e.g., zone ID) included in the assistance information. For instance, a UE **115** may know what zone it is in (e.g., zone **215** or zone **220**) where a zone is a geographical area with a corresponding zone ID. From the change rate of zones (e.g., zone IDs), a moving speed can be determined at UE **115**-*c* or **115**-*d* over time. For instance, the size of each zone may be known, and based on the location of a UE **115** in different zones over time, the speed of the UE **115** may be determined.

(82) At **410**, UE **115**-*c* may optionally determine a relative speed between itself, UE **115**-*c*, and UE **115**-*d*. In some cases, this determination may be based on the received transmitter assistance information. For example, if UE **115**-*c* is in speed category low in a first direction, and UE **115**-*d* is in speed category low in a second direction, and if the first and second directions are opposite, then the relative speed is determined as within the medium speed category if a three category system is used (e.g., FIGS. **3**C through **3**E).

(83) At **415**, UE **115**-*c* may optionally determine a DMRS pattern and corresponding index based on the determination of relative speed at **410**. Example DMRS patterns are shown in FIGS. **3**A-**3**E.

(84) At **420**, UE **115**-*c* may transmit receiver assistance information to UE **115**-*d*, which may be similar to transmitter assistance information. In some cases, this assistance information may include UE's **115**-*c* speed, direction, location (e.g., zone ID), relative speed, subcarrier spacing, and/or DMRS pattern index. The assistance information may include a DMRS pattern index based on the determination of a DMRS pattern at **415**. In some examples, assistance information may be transmitted in SCI, a sidelink data channel (SL-SCH) (e.g., MAC CE), or a feedback channel. Assistance information may be transmitted periodically (e.g., according to a schedule), after a change of assistance information (e.g., change in speed and/or direction of UE **115**-*a* or **115**-*b*), or based on a trigger from a transmitting UE **115**-*b*. More specifically, a change of assistance information at UE **115**-*c* may include when the change of speed exceeds a threshold (e.g., a change of absolute value is larger than a threshold, or change from one speed category to another: low to medium), a change of direction exceeds a threshold, the location is changed (e.g., zone ID changed), the change of relative speed exceeds a threshold (if UE **115**-*c* determines relative speed of UEs **115**-*c* and **115**-*d*), and/or the determined DMRS pattern is changed (if UE **115**-*c* determines DMRS pattern).

(85) At **425**, UE **115**-*d* may optionally determine a relative speed between itself, UE **115**-*d*, and UE **115**-*d*. In some cases, this determination may be based on the received receiver assistance information, as discussed above in relation to the determination at **410**. For example, if UE **115**-*c* is in speed category medium in a first direction, and UE **115**-*d* is in speed category low in the same direction, then the relative speed is determined to be within the low speed category if a three category system is used (e.g., FIGS. **3**C through **3**E).

(86) At **430**, UE **115**-*d* may determine a DMRS pattern based on the receiver assistance information and optionally the determination or relative speed at **425**. Example DMRS patterns are shown in FIGS. **3**A-**3**E. UE **115**-*d* may maintain record of UE's **115**-*c* past assistance information, and UE **115**-*d* may redetermine a DMRS pattern when the assistance information changes. For example, UE **115**-*d* may maintain ten receivers' speed information and determined the relative speeds for the ten sidelinks. UE **115**-*d* may determine a DMRS pattern based on the relative speeds of the ten sidelinks, and UE **115**-*d* continues using the DMRS pattern if no relative speed change (or the change is not large enough to trigger a change of DMRS pattern).

(87) At **435**, UE **115**-*d* may transmit, to UE **115**-*c*, using the DMRS pattern determined at **430** along with data. In some cases, an indication of the DMRS pattern to be used at **435** may be sent to

UE **115**-*c*.

(88) FIG. **5** illustrates an example of a process flow **500** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. In some examples, process flow **500** may implement aspects of wireless communications system **100**. Process flow **500** may be implemented by a UE **115**-*e* and UE **115**-*f*, or any other examples of UEs **115** as described herein. Alternative examples of the following may be implemented, where some steps are performed in a different order than described or are not performed at all. In some cases, steps may include additional features not mentioned below, or further steps may be added.

(89) At **505**, a UE **115**-*f* may determine the speed (and optionally direction) of itself, UE **115**-*f* UE **115**-*f* may operate in a broadcast configuration and may not receive transmissions from the receiver UE **115**-*e*, and thus, UE **115**-*f* may not be able to receive assistance information from UE **115**-*e*. In some examples, UE **115**-*e* may be unable to transmit assistance information to UE **115**-*f*. In some cases, UE **115**-*f* may assume UE **115**-*e* is stationary or moving in the opposite direction at a similar speed as UE **115**-*f*.

(90) At **510**, the UE **115**-*f* may determine a DMRS pattern for the sidelink communications with UE **115**-*e* based on the determined speed at **505**. Example DMRS patterns are shown in FIGS. **3**A-**3**E.

(91) At **515**, UE **115**-*f* may transmit, to UE **115**-*e*, using the DMRS pattern determined at **510** along with data. In some cases, an indication of the DMRS pattern to be used at **510** may be sent to UE **115**-*e* (e.g., via SCI).

(92) FIG. **6** illustrates an example of a process flow **600** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. In some examples, process flow **600** may implement aspects of wireless communications system **100**. Process flow **600** may be implemented by UE **115**-*g*, UE **115**-*e*, base station **105**-*b*, or any other examples of UEs **115** or base stations **105** as described herein. Alternative examples of the following may be implemented, where some steps are performed in a different order than described or are not performed at all. In some cases, steps may include additional features not mentioned below, or further steps may be added.

(93) At **602**, UE **115**-*g* may optionally transmit, to UE **115**-*h*, sidelink communications. For example, UE **115**-*g* may transmit data to UE **115**-*h* for V2X or V2V communications. In some cases, UEs **115**-*g* and **115**-*h* may be operating in mode 1 such that base station **105**-*b* assists in sidelink scheduling.

(94) At **605**, UE **115**-*g* may optionally transmit receiver assistance information to base station **105**-*b*. At **610**, UE **115**-*h* may transmit transmitter assistance information to base station **105**-*b*. In some cases, one or both of receiver and transmitter assistance information may be transmitted to base station **105**-*b*. In some cases, the assistance information may include each respective UE's **115**-*g* or **115**-*h* speed, direction, location (e.g., zone ID), relative speed, subcarrier spacing, and/or DMRS pattern index. In some examples, assistance information may be transmitted in UCI, a data channel (e.g., PUSCH, MAC CE), or a feedback channel.

(95) At **615**, base station **105**-*b* may determine a relative speed between UE **115**-*g* and UE **115**-*h*. In some cases, this determination at **615** may be based on the received assistance information (e.g., receiver and/or transmitter assistance information), similar to the relative speed determinations in process flow **400**.

(96) At **620**, base station **105**-*b* may determine a DMRS pattern and corresponding DMRS pattern index based on the determination of relative speed at **615**. At **625**, base station **105**-*b* may transmit, to UE **115**-*h*, the DMRS pattern index based on the determination of a DMRS pattern at **620**. Example DMRS patterns are shown in FIGS. **3**A-**3**E.

(97) At **630**, UE **115**-*h* may determine the DMRS pattern based on the received DMRS pattern index at **625** (e.g., based on the known mapping configuration). For example, two speed categories

and DMRS patterns may be configured, such as a low speed category (e.g., 0 to 120 kmph) may correspond to a low density DMRS pattern, while a high speed category (e.g., greater than 120 kmph) may correspond to a high density DMRS pattern. The mapping may be predefined (e.g., a table is stored at the UEs **115**-*g* and **115**-*h*), preconfigured (multiple tables may be stored at the UEs **115**-*g* and **115**-*h*), or configured by higher layer signaling (e.g., RRC signaling from base station **105**-*b*).
(98) At **635**, UE **115**-*h* may transmit, to UE **115**-*g*, using the DMRS pattern determined at **620** and **630** along with data. In some cases, an indication of the DMRS pattern to be used at **635** may be sent to UE **115**-*g*.
(99) FIG. **7** illustrates an example of a process flow **700** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. In some examples, process flow **700** may implement aspects of wireless communications system **100**. Process flow **700** may be implemented by a UE **115**-*i* and UE **115**-*j*, or any other examples of UEs **115** as described herein. Alternative examples of the following may be implemented, where some steps are performed in a different order than described or are not performed at all. In some cases, steps may include additional features not mentioned below, or further steps may be added.
(100) At **705**, UE **115**-*i* may receive a sidelink communication (e.g., data transmission) from UE **115**-*j*. At **710**, UE **115**-*i* may transmit, to UE **115**-*j*, HARQ feedback (e.g., acknowledgment (ACK) or NACK) for the sidelink communication received at **705**.
(101) At **715**, UE **115**-*j* may optionally determine an error based on the received HARQ feedback. In some cases, this error determination may be compared to a threshold and tracked over time, and UE **115**-*j* may reselect a DMRS pattern when the error rate has changed. For example, UE **115**-*j* may compute the BLER or packet error from HARQ feedback **710** over a time period, and if the BLER is higher than a target, the UE **115**-*j* may determine that a DMRS pattern with denser DMRS symbol locations should be used in subsequent sidelink communications. Error determinations may apply to unicast configurations.
(102) At **720**, UE **115**-*j* may optionally determine received power of the HARQ feedback (e.g., NACK when in a groupcast configuration with SFN feedback). In groupcast, only NACK(s) may be received by UE **115**-*j*. In some cases, this received power may be compared to a threshold and tracked over time, and UE **115**-*j* may reselect a DMRS pattern when the received power has changed. For example, UE **115**-*j* may measure the received power of a NACK signal in HARQ feedback **710**, and if the power is higher than a threshold, the UE **115**-*j* may determine that a DMRS pattern with denser DMRS symbol locations should be used in subsequent sidelink communications. The received power measurement may be a one-shot measurement or measurement over multiple NACK receptions.
(103) At **725**, UE **115**-*j* may determine (or redetermine) a DMRS pattern based on one or both of the comparisons of error and received power with their respective thresholds.
(104) At **730**, UE **115**-*j* may transmit, to UE **115**-*i*, using the DMRS pattern determined at **725** along with data. In some cases, an indication of the DMRS pattern to be used at **730** may be sent to UE **115**-*i*.
(105) FIG. **8** shows a block diagram **800** of a device **805** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The device **805** may be an example of aspects of a UE **115** as described herein. The device **805** may include a receiver **810**, a reference signal manager **815**, and a transmitter **820**. The device **805** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).
(106) The receiver **810** may receive information such as packets, user data, or control information associated with various information channels (e.g., control channels, data channels, and information related to reference signal patterns based on relative speed between a transmitter and

receiver, etc.). Information may be passed on to other components of the device **805**. The receiver **810** may be an example of aspects of the transceiver **1120** described with reference to FIG. **11**. The receiver **810** may utilize a single antenna or a set of antennas.
(107) The reference signal manager **815** may transmit assistance information to a second wireless device, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof and determine an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern based on the assistance information. The reference signal manager **815** may also determine a speed of the first wireless device, identify a subcarrier spacing for communicating with one or more other wireless devices, determine, based on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data, and transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern. The reference signal manager **815** may also transmit assistance information to a base station, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof, receive, from the base station, an indication of a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the assistance information, and transmit the data and the reference signal to a second wireless device, the reference signal being transmitted in accordance with the reference signal pattern. The reference signal manager **815** may also receive, from one or more other wireless devices, feedback information in response to one or more previous data transmissions to the one or more other wireless devices, determine a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the feedback information, and transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern. The reference signal manager **815** may be an example of aspects of the reference signal manager **1110** described herein.
(108) The actions performed by the reference signal manager **815** as described herein may be implemented to realize one or more potential advantages. One implementation may allow a UE **115** to save power and increase battery life by avoiding unsuccessful reception or transmission of communications when traveling at different speeds by optimizing the DMRS density for the current speed of the UE. Additionally or alternatively, the UE **115** may further reduce signaling overhead by reducing the DMRS density at slow speeds when dense DMRS is not needed for successful communications. Another implementation may provide improved quality and reliability of service at the UE **115**, as latency and the number of separate resources allocated to the UE **115** may be optimized based on the DMRS density, which may be based on the speed or location of UE **115**.
(109) The reference signal manager **815**, or its sub-components, may be implemented in hardware, code (e.g., software or firmware) executed by a processor, or any combination thereof. If implemented in code executed by a processor, the functions of the reference signal manager **815**, or its sub-components may be executed by a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described in the present disclosure.
(110) The reference signal manager **815**, or its sub-components, may be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations by one or more physical components. In some examples, the reference signal manager **815**, or its sub-components, may be a separate and distinct component in accordance with various aspects of the present disclosure. In some examples, the reference signal manager **815**, or its sub-components, may be combined with one or more other hardware components, including but not limited to an input/output (I/O) component, a transceiver, a network

server, another computing device, one or more other components described in the present disclosure, or a combination thereof in accordance with various aspects of the present disclosure.
(111) The transmitter **820** may transmit signals generated by other components of the device **805**. In some examples, the transmitter **820** may be collocated with a receiver **810** in a transceiver module. For example, the transmitter **820** may be an example of aspects of the transceiver **1120** described with reference to FIG. **11**. The transmitter **820** may utilize a single antenna or a set of antennas.
(112) FIG. **9** shows a block diagram **900** of a device **905** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The device **905** may be an example of aspects of a device **805**, or a UE **115** as described herein. The device **905** may include a receiver **910**, a reference signal manager **915**, and a transmitter **950**. The device **905** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).
(113) The receiver **910** may receive information such as packets, user data, or control information associated with various information channels (e.g., control channels, data channels, and information related to reference signal patterns based on relative speed between a transmitter and receiver, etc.). Information may be passed on to other components of the device **905**. The receiver **910** may be an example of aspects of the transceiver **1120** described with reference to FIG. **11**. The receiver **910** may utilize a single antenna or a set of antennas.
(114) The reference signal manager **915** may be an example of aspects of the reference signal manager **815** as described herein. The reference signal manager **915** may include a mobility manager **920**, a reference signal pattern manager **925**, a speed component **930**, a subcarrier spacing component **935**, a communications manager **940**, and a feedback component **945**. The reference signal manager **915** may be an example of aspects of the reference signal manager **1110** described herein.
(115) The mobility manager **920** may transmit assistance information to a second wireless device, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof.
(116) The reference signal pattern manager **925** may determine an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern based on the assistance information. The speed component **930** may determine a speed of the first wireless device. The subcarrier spacing component **935** may identify a subcarrier spacing for communicating with one or more other wireless devices.
(117) The reference signal pattern manager **925** may determine, based on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data. The communications manager **940** may transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(118) The mobility manager **920** may transmit assistance information to a base station, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof. The reference signal pattern manager **925** may receive, from the base station, an indication of a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the assistance information.
(119) The communications manager **940** may transmit the data and the reference signal to a second wireless device, the reference signal being transmitted in accordance with the reference signal pattern. The feedback component **945** may receive, from one or more other wireless devices, feedback information in response to one or more previous data transmissions to the one or more other wireless devices.
(120) The reference signal pattern manager **925** may determine a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the feedback

information. The communications manager **940** may transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.

(121) The transmitter **950** may transmit signals generated by other components of the device **905**. In some examples, the transmitter **950** may be collocated with a receiver **910** in a transceiver module. For example, the transmitter **950** may be an example of aspects of the transceiver **1120** described with reference to FIG. **11**. The transmitter **950** may utilize a single antenna or a set of antennas.

(122) FIG. **10** shows a block diagram **1000** of a reference signal manager **1005** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The reference signal manager **1005** may be an example of aspects of a reference signal manager **815**, a reference signal manager **915**, or a reference signal manager **1110** described herein. The reference signal manager **1005** may include a mobility manager **1010**, a reference signal pattern manager **1015**, a speed component **1020**, a direction component **1025**, a location component **1030**, a relative speed manager **1035**, a communications manager **1040**, a subcarrier spacing component **1045**, memory **1050**, and a feedback component **1055**. Each of these modules may communicate, directly or indirectly, with one another (e.g., via one or more buses).

(123) The mobility manager **1010** may transmit assistance information to a second wireless device, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof.

(124) In some examples, the mobility manager **1010** may transmit assistance information to a base station, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof. In some examples, the mobility manager **1010** may transmit, as part of the assistance information, a speed value indicating the speed of the first wireless device. In some examples, the mobility manager **1010** may transmit, as part of the assistance information, a direction value indicating the movement direction of the first wireless device.

(125) In some examples, the mobility manager **1010** may transmit, as part of the assistance information, a location value indicating the location of the first wireless device. In some examples, the mobility manager **1010** may determine a velocity of the first wireless device. In some examples, the mobility manager **1010** may transmit, as part of the assistance information, an indication of the relative speed.

(126) In some examples, the mobility manager **1010** may transmit, as part of the assistance information, an indication of the reference signal pattern. In some examples, the mobility manager **1010** may identify a highest relative speed from the relative speeds of the set of communication links, where the reference signal pattern is based on the highest relative speed.

(127) In some examples, the mobility manager **1010** may receive first assistance information and second assistance information from the second wireless device, each of the first assistance information and the second assistance information being associated with a respective speed of the second wireless device, or a respective location of the second wireless device, or a combination thereof. In some examples, the mobility manager **1010** may identify a periodicity for transmitting the assistance information, where the assistance information is transmitted in accordance with the periodicity.

(128) In some examples, the mobility manager **1010** may generate first assistance information and second assistance information for the first wireless device, each of the first assistance information and the second assistance information being associated with a respective speed of the first wireless device, or a respective location of the first wireless device, or a combination thereof.

(129) In some examples, the mobility manager **1010** may identify a difference between the first assistance information and the second assistance information. In some examples, the mobility

manager **1010** may transmit the assistance information based on identifying the difference. In some examples, the mobility manager **1010** may receive a request to transmit the assistance information, where the assistance information is transmitted in response to the received request.

(130) In some examples, the mobility manager **1010** may transmit the assistance information as part of sidelink control information, or as part of a medium access control (MAC) control element, or over a sidelink data channel, or over a sidelink shared channel, or over a feedback channel, or a combination thereof. The reference signal pattern manager **1015** may identify a reference signal pattern of a reference signal for demodulating data, the reference signal pattern based on the assistance information.

(131) In some examples, the reference signal pattern manager **1015** may determine, based on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data. In some examples, the reference signal pattern manager **1015** may receive, from the base station, an indication of a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the assistance information.

(132) In some examples, the reference signal pattern manager **1015** may determine a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the feedback information. In some examples, the reference signal pattern manager **1015** may determine the reference signal pattern based on the relative speed.

(133) In some examples, the reference signal pattern manager **1015** may receive, from the second wireless device, an indication of the reference signal pattern. In some examples, the reference signal pattern manager **1015** may determine the reference signal pattern based on the indication. In some examples, the reference signal pattern manager **1015** may receive, from a set of wireless devices, respective indications of one or more reference signal patterns.

(134) In some examples, the reference signal pattern manager **1015** may determine the reference signal pattern based on the one or more reference signal patterns. In some examples, the reference signal pattern manager **1015** may select the reference signal pattern from one or more reference signal patterns, each of the one or more reference signal patterns corresponding to a respective relative speed, or a range of relative speeds, or a combination thereof.

(135) In some examples, the reference signal pattern manager **1015** may identify a configuration of a map between the each of the one or more reference signal patterns and the respective relative speed, or the range of relative speeds, or a combination thereof. In some examples, the reference signal pattern manager **1015** may determine the reference signal pattern based on the relative speeds of the set of communication links. In some examples, the reference signal pattern manager **1015** may determine the reference signal pattern based on the subcarrier spacing and a relative speed between the first wireless device and the second wireless device.

(136) In some examples, the reference signal pattern manager **1015** may identify a difference between the first assistance information and the second assistance information. In some examples, the reference signal pattern manager **1015** may determine the reference signal pattern based on the difference. In some examples, the reference signal pattern manager **1015** may transmit an indication of the reference signal pattern to the one or more other wireless devices.

(137) In some examples, the reference signal pattern manager **1015** may determine the reference signal pattern based on the error rate satisfying an error rate threshold. In some examples, the reference signal pattern manager **1015** may determine the reference signal pattern based on the received power satisfying a power threshold. In some cases, the reference signal pattern is based on a map between the one or more reference signal patterns and a respective relative speed, or a range of relative speeds, or a combination thereof, and where the reference signal pattern is determined based on the map and the subcarrier spacing.

(138) In some cases, the reference signal pattern indicates one or more symbol periods during which the reference signal is transmitted and a gap between each of the one or more symbol

periods. In some cases, the reference signal pattern is from a set of reference signal patterns, each reference signal pattern of the set including a different number of symbol periods within a slot that include the reference signal.
(139) In some cases, the reference signal pattern is from a set of reference signal patterns, each reference signal pattern of the set including a different gap between symbol periods within a slot that include the reference signal. In some cases, the indication is transmitted within control information to the one or more other wireless devices.
(140) In some cases, the reference signal pattern is from one or more reference signal patterns, each of the one or more reference signal patterns corresponding to a respective speed, or a range of speeds, or a combination thereof. In some cases, the reference signal pattern includes one or more symbol periods during which the reference signal is transmitted and a gap between each of the one or more symbol periods.
(141) The speed component **1020** may determine a speed of the first wireless device. In some examples, the speed component **1020** may determine the speed of the first wireless device. In some examples, the speed component **1020** may determine an absolute speed of the first wireless device. In some examples, the speed component **1020** may identify the speed value from a speed index based on the absolute speed, the speed index including a map between respective speed values and one or more absolute speeds, or a range of absolute speeds, or a combination thereof.
(142) In some examples, the speed component **1020** may identify a configuration of the speed index, where identifying the speed value is based on the configuration. In some examples, the speed component **1020** may receive an indication of the configuration of the speed index. In some examples, the speed component **1020** may receive, from the second wireless device, an indication of a speed of the second wireless device, or a velocity of the second wireless device, or a location of the second wireless device, or a combination thereof.
(143) In some examples, the speed component **1020** may calculate the speed of the second wireless device based on the one or more zones the second wireless device was located in during the time period. In some cases, the speed index is from a set of speed indices. The communications manager **1040** may transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(144) In some examples, the communications manager **1040** may transmit the data and the reference signal to a second wireless device, the reference signal being transmitted in accordance with the reference signal pattern. In some examples, the communications manager **1040** may transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(145) In some examples, the communications manager **1040** may identify a set of communication links with other wireless devices, each communication link of the set of communication links being associated with a relative speed between the first wireless device and a respective wireless device. In some examples, the communications manager **1040** may determine an error rate threshold for transmitting the data over the set of communication links, where the reference signal pattern is based on the relative speeds of the set of communication links and the error rate threshold.
(146) In some examples, the communications manager **1040** may transmit the data and the reference signal to the second wireless device, the reference signal being transmitted in accordance with the reference signal pattern. In some examples, the communications manager **1040** may receive the data and the reference signal from the second wireless device, the reference signal being received in accordance with the reference signal pattern.
(147) In some examples, communicating with the second wireless device using sidelink communications, where the first wireless device includes a first UE and the second wireless device includes a second UE. In some examples, the communications manager **1040** may broadcast the data and the reference signal to the one or more other wireless devices.
(148) In some examples, operating under a mode for sidelink communications, where the first

wireless device includes a first UE and the second wireless device includes a second UE.
(149) The subcarrier spacing component **1045** may identify a subcarrier spacing for communicating with one or more other wireless devices. In some examples, the subcarrier spacing component **1045** may identify a subcarrier spacing for communicating with the second wireless device.
(150) The feedback component **1055** may receive, from one or more other wireless devices, feedback information in response to one or more previous data transmissions to the one or more other wireless devices. In some examples, the feedback component **1055** may compute an error rate over a time period based on the feedback information. In some examples, the feedback component **1055** may measure a received power of the feedback information.
(151) In some cases, the error rate includes a block error rate or a packet error rate of the one or more previous data transmissions. In some cases, the feedback information includes one or more negative acknowledgments received during a time period. The direction component **1025** may determine a movement direction of the first wireless device.
(152) In some examples, the direction component **1025** may determine an absolute movement direction of the first wireless device. In some examples, the direction component **1025** may identify the direction value from a direction index based on the absolute movement direction, the direction index including a map between one or more absolute movement directions and respective direction values.
(153) In some cases, each of the respective direction values are based on an angular offset from a cardinal direction or an intercardinal direction. The location component **1030** may determine the location of the first wireless device. In some examples, identifying a zone identifier from a set of zone identifiers based on the location of the first wireless device, each of the set of zone identifiers being associated with a respective geographic area, where the location value includes the zone identifier.
(154) In some examples, determining, based on the location of the second wireless device, one or more zones the second wireless device was located in during a time period, where the received indication includes one or more zone identifiers corresponding to respective zones.
(155) The relative speed manager **1035** may identify a relative speed between the first wireless device and the second wireless device based on the assistance information, where the reference signal pattern is determined based on the relative speed.
(156) In some examples, the relative speed manager **1035** may determine the relative speed based on the received indication. In some examples, the relative speed manager **1035** may calculate the relative speed based on the speed of the first wireless device and the speed of the second wireless device. In some examples, the relative speed manager **1035** may calculate the relative speed based on the velocity of the first wireless device and the velocity of the second wireless device. The memory **1050** may store the first assistance information and the second assistance information at the first wireless device.
(157) FIG. **11** shows a diagram of a system **1100** including a device **1105** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The device **1105** may be an example of or include the components of device **805**, device **905**, or a UE **115** as described herein. The device **1105** may include components for bi-directional voice and data communications including components for transmitting and receiving communications, including a reference signal manager **1110**, an I/O controller **1115**, a transceiver **1120**, an antenna **1125**, memory **1130**, and a processor **1140**. These components may be in electronic communication via one or more buses (e.g., bus **1145**).
(158) The reference signal manager **1110** may transmit assistance information to a second wireless device, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof and identify a reference signal pattern of a reference signal for demodulating data, the reference signal pattern based on the assistance information. The reference signal manager **1110** may also determine a speed of the first wireless

device, identify a subcarrier spacing for communicating with one or more other wireless devices, determine, based on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data, and transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern. The reference signal manager **1110** may also transmit assistance information to a base station, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof, receive, from the base station, an indication of a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the assistance information, and transmit the data and the reference signal to a second wireless device, the reference signal being transmitted in accordance with the reference signal pattern. The reference signal manager **1110** may also receive, from one or more other wireless devices, feedback information in response to one or more previous data transmissions to the one or more other wireless devices, determine a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the feedback information, and transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.

(159) The I/O controller **1115** may manage input and output signals for the device **1105**. The I/O controller **1115** may also manage peripherals not integrated into the device **1105**. In some cases, the I/O controller **1115** may represent a physical connection or port to an external peripheral. In some cases, the I/O controller **1115** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. In other cases, the I/O controller **1115** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller **1115** may be implemented as part of a processor. In some cases, a user may interact with the device **1105** via the I/O controller **1115** or via hardware components controlled by the I/O controller **1115**.

(160) The transceiver **1120** may communicate bi-directionally, via one or more antennas, wired, or wireless links as described herein. For example, the transceiver **1120** may represent a wireless transceiver and may communicate bi-directionally with another wireless transceiver. The transceiver **1120** may also include a modem to modulate the packets and provide the modulated packets to the antennas for transmission, and to demodulate packets received from the antennas.

(161) In some cases, the wireless device may include a single antenna **1125**. However, in some cases, the device may have more than one antenna **1125**, which may be capable of concurrently transmitting or receiving multiple wireless transmissions.

(162) The memory **1130** may include random-access memory (RAM) and read-only memory (ROM). The memory **1130** may store computer-readable, computer-executable code **1135** including instructions that, when executed, cause the processor to perform various functions described herein. In some cases, the memory **1130** may contain, among other things, a basic input/output system (BIOS) which may control basic hardware or software operation such as the interaction with peripheral components or devices.

(163) The processor **1140** may include an intelligent hardware device, (e.g., a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, an FPGA, a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some cases, the processor **1140** may be configured to operate a memory array using a memory controller. In other cases, a memory controller may be integrated into the processor **1140**. The processor **1140** may be configured to execute computer-readable instructions stored in a memory (e.g., the memory **1130**) to cause the device **1105** to perform various functions (e.g., functions or tasks supporting reference signal patterns based on relative speed between a transmitter and receiver).

(164) The code **1135** may include instructions to implement aspects of the present disclosure,

including instructions to support wireless communications. The code **1135** may be stored in a non-transitory computer-readable medium such as system memory or other type of memory. In some cases, the code **1135** may not be directly executable by the processor **1140** but may cause a computer (e.g., when compiled and executed) to perform functions described herein.
(165) FIG. **12** shows a flowchart illustrating a method **1200** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The operations of method **1200** may be implemented by a UE **115** or its components as described herein. For example, the operations of method **1200** may be performed by a reference signal manager as described with reference to FIGS. **8** through **11**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the functions described herein. Additionally or alternatively, a UE may perform aspects of the functions described herein using special-purpose hardware.
(166) At **1205**, the UE may transmit assistance information to a second wireless device. For example, the assistance information may be associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof. The operations of **1205** may be performed according to the methods described herein. In some examples, aspects of the operations of **1205** may be performed by a mobility manager as described with reference to FIGS. **8** through **11**.
(167) At **1210**, the UE may determine an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern based on the assistance information. The operations of **1210** may be performed according to the methods described herein. In some examples, aspects of the operations of **1210** may be performed by a reference signal pattern manager as described with reference to FIGS. **8** through **11**.
(168) FIG. **13** shows a flowchart illustrating a method **1300** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The operations of method **1300** may be implemented by a UE **115** or its components as described herein. For example, the operations of method **1300** may be performed by a reference signal manager as described with reference to FIGS. **8** through **11**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the functions described herein. Additionally or alternatively, a UE may perform aspects of the functions described herein using special-purpose hardware.
(169) At **1305**, the UE may transmit assistance information to a second wireless device. For example, the assistance information may be associated with a speed of the first wireless device (e.g., the UE), or a location of the first wireless device, or a combination thereof. The operations of **1305** may be performed according to the methods described herein. In some examples, aspects of the operations of **1305** may be performed by a mobility manager as described with reference to FIGS. **8** through **11**.
(170) At **1310**, the UE may identify a relative speed between the first wireless device and the second wireless device based on the assistance information. The operations of **1310** may be performed according to the methods described herein. In some examples, aspects of the operations of **1310** may be performed by a relative speed manager as described with reference to FIGS. **8** through **11**.
(171) At **1315**, the UE may determine an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern based on the assistance information and the relative speed. The operations of **1315** may be performed according to the methods described herein. In some examples, aspects of the operations of **1315** may be performed by a reference signal pattern manager as described with reference to FIGS. **8** through **11**.
(172) FIG. **14** shows a flowchart illustrating a method **1400** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The operations of method **1400** may be implemented by a UE **115** or its

components as described herein. For example, the operations of method **1400** may be performed by a reference signal manager as described with reference to FIGS. **8** through **11**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the functions described herein. Additionally or alternatively, a UE may perform aspects of the functions described herein using special-purpose hardware.

(173) At **1405**, the UE may determine the speed of a first wireless device (e.g., the UE). The operations of **1405** may be performed according to the methods described herein. In some examples, aspects of the operations of **1405** may be performed by a speed component as described with reference to FIGS. **8** through **11**.

(174) At **1410**, the UE may determine a movement direction of the first wireless device. The operations of **1410** may be performed according to the methods described herein. In some examples, aspects of the operations of **1410** may be performed by a direction component as described with reference to FIGS. **8** through **11**.

(175) At **1415**, the UE may transmit assistance information to a second wireless device (e.g., another UE), the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof. The operations of **1415** may be performed according to the methods described herein. In some examples, aspects of the operations of **1415** may be performed by a mobility manager as described with reference to FIGS. **8** through **11**.

(176) At **1420**, the UE may transmit, as part of the assistance information, a speed value indicating the speed of the first wireless device. The operations of **1420** may be performed according to the methods described herein. In some examples, aspects of the operations of **1420** may be performed by a mobility manager as described with reference to FIGS. **8** through **11**.

(177) At **1425**, the UE may also transmit, as part of the assistance information, a direction value indicating the movement direction of the first wireless device. That is, the UE may options transmit speed information, or both speed and movement direction information within the assistance information. The operations of **1425** may be performed according to the methods described herein. In some examples, aspects of the operations of **1425** may be performed by a mobility manager as described with reference to FIGS. **8** through **11**.

(178) At **1430**, the UE may determine an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern based on the assistance information. The operations of **1430** may be performed according to the methods described herein. In some examples, aspects of the operations of **1430** may be performed by a reference signal pattern manager as described with reference to FIGS. **8** through **11**.

(179) FIG. **15** shows a flowchart illustrating a method **1500** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The operations of method **1500** may be implemented by a UE **115** or its components as described herein. For example, the operations of method **1500** may be performed by a reference signal manager as described with reference to FIGS. **8** through **11**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the functions described herein. Additionally or alternatively, a UE may perform aspects of the functions described herein using special-purpose hardware.

(180) At **1505**, the UE may determine the location of a first wireless device (e.g., the UE). The operations of **1505** may be performed according to the methods described herein. In some examples, aspects of the operations of **1505** may be performed by a location component as described with reference to FIGS. **8** through **11**.

(181) At **1510**, the UE may transmit assistance information to a second wireless device, the assistance information associated with a speed of the first wireless device, or a location of the first wireless device, or a combination thereof. The operations of **1510** may be performed according to the methods described herein. In some examples, aspects of the operations of **1510** may be

performed by a mobility manager as described with reference to FIGS. **8** through **11**.
(182) At **1515**, the UE may transmit, as part of the assistance information, a location value indicating the location of the first wireless device. The operations of **1515** may be performed according to the methods described herein. In some examples, aspects of the operations of **1515** may be performed by a mobility manager as described with reference to FIGS. **8** through **11**.
(183) At **1520**, the UE may determine an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern based on the assistance information. The operations of **1520** may be performed according to the methods described herein. In some examples, aspects of the operations of **1520** may be performed by a reference signal pattern manager as described with reference to FIGS. **8** through **11**.
(184) FIG. **16** shows a flowchart illustrating a method **1600** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The operations of method **1600** may be implemented by a UE **115** or its components as described herein. For example, the operations of method **1600** may be performed by a reference signal manager as described with reference to FIGS. **8** through **11**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the functions described herein. Additionally or alternatively, a UE may perform aspects of the functions described herein using special-purpose hardware.
(185) At **1605**, the UE may determine a speed of the first wireless device (e.g., the UE). The operations of **1605** may be performed according to the methods described herein. In some examples, aspects of the operations of **1605** may be performed by a speed component as described with reference to FIGS. **8** through **11**.
(186) At **1610**, the UE may identify a subcarrier spacing for communicating with one or more other wireless devices. The operations of **1610** may be performed according to the methods described herein. In some examples, aspects of the operations of **1610** may be performed by a subcarrier spacing component as described with reference to FIGS. **8** through **11**.
(187) At **1615**, the UE may determine, based on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data. The operations of **1615** may be performed according to the methods described herein. In some examples, aspects of the operations of **1615** may be performed by a reference signal pattern manager as described with reference to FIGS. **8** through **11**.
(188) At **1620**, the UE may transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern. The operations of **1620** may be performed according to the methods described herein. In some examples, aspects of the operations of **1620** may be performed by a communications manager as described with reference to FIGS. **8** through **11**.
(189) FIG. **17** shows a flowchart illustrating a method **1700** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The operations of method **1700** may be implemented by a UE **115** or its components as described herein. For example, the operations of method **1700** may be performed by a reference signal manager as described with reference to FIGS. **8** through **11**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the functions described herein. Additionally or alternatively, a UE may perform aspects of the functions described herein using special-purpose hardware.
(190) At **1705**, the UE may transmit assistance information to a base station. For example, the assistance information may be associated with a speed of the first wireless device (e.g., the UE), or a location of the first wireless device, or a combination thereof. The operations of **1705** may be performed according to the methods described herein. In some examples, aspects of the operations of **1705** may be performed by a mobility manager as described with reference to FIGS. **8** through **11**.

(191) At **1710**, the UE may receive, from the base station, an indication of an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern being based on the assistance information. The operations of **1710** may be performed according to the methods described herein. In some examples, aspects of the operations of **1710** may be performed by a reference signal pattern manager as described with reference to FIGS. **8** through **11**.
(192) At **1715**, the UE may transmit the data and the reference signal to a second wireless device, the reference signal being transmitted in accordance with the adaptive reference signal pattern. The operations of **1715** may be performed according to the methods described herein. In some examples, aspects of the operations of **1715** may be performed by a communications manager as described with reference to FIGS. **8** through **11**.
(193) FIG. **18** shows a flowchart illustrating a method **1800** that supports reference signal patterns based on relative speed between a transmitter and receiver in accordance with one or more aspects of the present disclosure. The operations of method **1800** may be implemented by a UE **115** or its components as described herein. For example, the operations of method **1800** may be performed by a reference signal manager as described with reference to FIGS. **8** through **11**. In some examples, a UE may execute a set of instructions to control the functional elements of the UE to perform the functions described herein. Additionally or alternatively, a UE may perform aspects of the functions described herein using special-purpose hardware.
(194) At **1805**, the UE may receive, from one or more other wireless devices, feedback information in response to one or more previous data transmissions to the one or more other wireless devices. The operations of **1805** may be performed according to the methods described herein. In some examples, aspects of the operations of **1805** may be performed by a feedback component as described with reference to FIGS. **8** through **11**.
(195) At **1810**, the UE may determine a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based on the feedback information. The operations of **1810** may be performed according to the methods described herein. In some examples, aspects of the operations of **1810** may be performed by a reference signal pattern manager as described with reference to FIGS. **8** through **11**.
(196) At **1815**, the UE may transmit the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern. The operations of **1815** may be performed according to the methods described herein. In some examples, aspects of the operations of **1815** may be performed by a communications manager as described with reference to FIGS. **8** through **11**.
(197) The following provides an overview of examples of the present disclosure:
(198) Example 1: A method for wireless communications at a first wireless device comprising: transmitting assistance information to a second wireless device; and determining an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern based at least in part on the assistance information.
(199) Example 2: The method of example 1, further comprising: determining a speed of the first wireless device; and transmitting, as part of the assistance information, a speed value indicating the speed of the first wireless device.
(200) Example 3: The method of any one of examples 1 and 2, further comprising: determining a movement direction of the first wireless device; and transmitting, as part of the assistance information, a direction value indicating the movement direction of the first wireless device.
(201) Example 4: The method of any one of examples 1 and 2, further comprising: determining an absolute speed of the first wireless device; and identifying the speed value from a speed index based at least in part on the absolute speed, the speed index comprising a map between respective speed values and one or more absolute speeds, or a range of absolute speeds, or a combination thereof.
(202) Example 5: The method of any one of examples 1 through 4, further comprising: determining

a location of the first wireless device; and transmitting, as part of the assistance information, a location value indicating the location of the first wireless device.

(203) Example 6: The method of any one of examples 1 through 5, further comprising: identifying a relative speed between the first wireless device and the second wireless device based at least in part on the assistance information, wherein the adaptive reference signal pattern is determined based at least in part on the relative speed.

(204) Example 7: The method of example 1 through 6, wherein identifying the relative speed comprises: receiving, from the second wireless device, an indication of a speed of the second wireless device, or a velocity of the second wireless device, or a location of the second wireless device, or a combination thereof; and determining the relative speed based at least in part on the indication.

(205) Example 8: The method of any one of examples 1 through 7, further comprising: transmitting, as part of the assistance information, an indication of the relative speed.

(206) Example 9: The method of any one of examples 1 through 8, further comprising: determining the adaptive reference signal pattern based at least in part on the relative speed; and transmitting, as part of the assistance information, an indication of the adaptive reference signal pattern.

(207) Example 10: The method of any one of examples 1 through 9, wherein determining the adaptive reference signal pattern comprises: receiving, from the second wireless device, an indication of the adaptive reference signal pattern; and determining the adaptive reference signal pattern based at least in part on the indication.

(208) Example 11: The method of any one of examples 1 through 10, wherein determining the adaptive reference signal pattern comprises: receiving, from a plurality of wireless devices, respective indications of one or more reference signal patterns; and determining the adaptive reference signal pattern based at least in part on the one or more reference signal patterns.

(209) Example 12: The method of any one of examples 1 through 11, wherein determining the adaptive reference signal pattern comprises: selecting the adaptive reference signal pattern from one or more reference signal patterns, each of the one or more reference signal patterns corresponding to a respective relative speed, or a range of relative speeds, or a combination thereof.

(210) Example 13: The method of any one of examples 1 through 12, wherein determining the adaptive reference signal pattern comprises: identifying a plurality of communication links with other wireless devices, each communication link of the plurality of communication links being associated with a relative speed between the first wireless device and a respective wireless device; and determining the adaptive reference signal pattern based at least in part on the relative speeds of the plurality of communication links.

(211) Example 14: The method of any one of examples 1 through 13, wherein determining the adaptive reference signal pattern comprises: identifying a subcarrier spacing for communicating with the second wireless device; and determining the adaptive reference signal pattern based at least in part on the subcarrier spacing and a relative speed between the first wireless device and the second wireless device.

(212) Example 15: The method of any one of examples 1 through 14, further comprising: receiving first assistance information and second assistance information from the second wireless device, each of the first assistance information and the second assistance information being associated with a respective speed of the second wireless device, or a respective location of the second wireless device, or a combination thereof; storing the first assistance information and the second assistance information at the first wireless device; identifying a difference between the first assistance information and the second assistance information; and determining the adaptive reference signal pattern based at least in part on the difference.

(213) Example 16: The method of any one of examples 1 through 15, further comprising: generating first assistance information and second assistance information for the first wireless device, each of the first assistance information and the second assistance information being

associated with a respective speed of the first wireless device, or a respective location of the first wireless device, or a combination thereof; identifying a difference between the first assistance information and the second assistance information; and transmitting the assistance information based at least in part on identifying the difference.
(214) Example 17: The method of any one of examples 1 through 16, further comprising: identifying a periodicity for transmitting the assistance information, wherein the assistance information is transmitted in accordance with the periodicity.
(215) Example 18: The method of any one of examples 1 through 17, further comprising: receiving a request to transmit the assistance information, wherein the assistance information is transmitted in response to the request.
(216) Example 19: The method of any one of examples 1 through 18, further comprising: transmitting the data and the reference signal to the second wireless device, the reference signal being transmitted in accordance with the adaptive reference signal pattern, wherein the adaptive reference signal pattern indicates one or more symbol periods during which the reference signal is transmitted and a gap between each of the one or more symbol periods.
(217) Example 20: The method of any one of examples 1 through 19, further comprising: receiving the data and the reference signal from the second wireless device, the reference signal being received in accordance with the adaptive reference signal pattern, wherein the adaptive reference signal pattern indicates one or more symbol periods during which the reference signal is transmitted and a gap between each of the one or more symbol periods.
(218) Example 21: The method of any one of examples 1 through 20, further comprising: communicating with the second wireless device using sidelink communications, wherein the first wireless device comprises a first user equipment (UE) and the second wireless device comprises a second UE.
(219) Example 22: The method of any one of examples 1 through 21, wherein the sidelink communications comprise vehicle to everything communications.
(220) Example 23: A method for wireless communications at a first wireless device, comprising: determining a speed of the first wireless device; identifying a subcarrier spacing for communicating with one or more other wireless devices; determining, based at least in part on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data; and transmitting the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(221) Example 24: The method of example 23, wherein transmitting the data and indication of the reference signal pattern comprises: broadcasting the data and the reference signal to the one or more other wireless devices.
(222) Example 25: The method of any one of examples 23 and 24, further comprising: transmitting an indication of the reference signal pattern to the one or more other wireless devices.
(223) Example 26: A method for wireless communications at a first wireless device comprising: transmitting assistance information to a base station; receiving, from the base station, an indication of an adaptive reference signal pattern of a reference signal for demodulating data, the adaptive reference signal pattern being based at least in part on the assistance information; and transmitting the data and the reference signal to a second wireless device, the reference signal being transmitted in accordance with the adaptive reference signal pattern.
(224) Example 27: The method of example 26, wherein the adaptive reference signal pattern is from one or more reference signal patterns, each of the one or more reference signal patterns corresponding to a respective speed, or a range of speeds, or a combination thereof.
(225) Example 28: The method of any one of examples 26 and 27, wherein the adaptive reference signal pattern comprises one or more symbol periods during which the reference signal is transmitted and a gap between each of the one or more symbol periods.

(226) Example 29: A method for wireless communications at a first wireless device, comprising: receiving, from one or more other wireless devices, feedback information in response to one or more previous data transmissions to the one or more other wireless devices; determining a reference signal pattern of a reference signal for demodulating data, the reference signal pattern being based at least in part on the feedback information; and transmitting the data and the reference signal to the one or more other wireless devices, the reference signal being transmitted in accordance with the reference signal pattern.
(227) Example 30: The method of example 29, wherein determining the reference signal pattern comprises: computing an error rate over a time period based at least in part on the feedback information; and determining the reference signal pattern based at least in part on the error rate satisfying an error rate threshold.
(228) Example 31: An apparatus for wireless communication comprising at least one means for performing a method of any one of examples 1 through 22.
(229) Example 32: An apparatus for wireless communication comprising a processor, and memory coupled to the processor, and the processor and memory may be configured to cause the apparatus to perform a method of any one of examples 1 through 22.
(230) Example 33: A non-transitory computer-readable medium storing code for wireless communication comprising a processor, memory in electronic communication with the processor, and instructions executable by the processor to cause the apparatus to perform a method of any one of examples 1 through 22.
(231) Example 34: An apparatus for wireless communication comprising at least one means for performing a method of any one of examples 23 through 25.
(232) Example 35: An apparatus for wireless communication comprising a processor, and memory coupled to the processor, and the processor and memory may be configured to cause the apparatus to perform a method of any one of examples 23 through 25.
(233) Example 36: A non-transitory computer-readable medium storing code for wireless communication comprising a processor, memory in electronic communication with the processor, and instructions executable by the processor to cause the apparatus to perform a method of any one of examples 23 through 25.
(234) Example 37: An apparatus for wireless communication comprising at least one means for performing a method of any one of examples 26 through 28.
(235) Example 38: An apparatus for wireless communication comprising a processor, and memory coupled to the processor, and the processor and memory may be configured to cause the apparatus to perform a method of any one of examples 26 through 28.
(236) Example 39: A non-transitory computer-readable medium storing code for wireless communication comprising a processor, memory in electronic communication with the processor, and instructions executable by the processor to cause the apparatus to perform a method of any one of examples 26 through 28.
(237) Example 40: An apparatus for wireless communication comprising at least one means for performing a method of any one of examples 29 and 30.
(238) Example 41: An apparatus for wireless communication comprising a processor, and memory coupled to the processor, and the processor and memory may be configured to cause the apparatus to perform a method of any one of examples 29 and 30.
(239) Example 42: A non-transitory computer-readable medium storing code for wireless communication comprising a processor, memory in electronic communication with the processor, and instructions executable by the processor to cause the apparatus to perform a method of any one of examples 29 and 30.
(240) It should be noted that the methods described herein describe possible implementations, and that the operations may be rearranged or otherwise modified and that other implementations are possible. Further, aspects from two or more of the methods may be combined.

(241) Techniques described herein may be used for various wireless communications systems such as code division multiple access (CDMA), time division multiple access (TDMA), frequency division multiple access (FDMA), orthogonal frequency division multiple access (OFDMA), single carrier frequency division multiple access (SC-FDMA), and other systems. A CDMA system may implement a radio technology such as CDMA2000, Universal Terrestrial Radio Access (UTRA), etc. CDMA2000 covers IS-2000, IS-95, and IS-856 standards. IS-2000 Releases may be commonly referred to as CDMA2000 1×, 1×, etc. IS-856 (TIA-856) is commonly referred to as CDMA2000 1×EV-DO, High Rate Packet Data (HRPD), etc. UTRA includes Wideband CDMA (WCDMA) and other variants of CDMA. A TDMA system may implement a radio technology such as Global System for Mobile Communications (GSM).

(242) An OFDMA system may implement a radio technology such as Ultra Mobile Broadband (UMB), Evolved UTRA (E-UTRA), Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM, etc. UTRA and E-UTRA are part of Universal Mobile Telecommunications System (UMTS). LTE, LTE-A, and LTE-A Pro are releases of UMTS that use E-UTRA. UTRA, E-UTRA, UMTS, LTE, LTE-A, LTE-A Pro, NR, and GSM are described in documents from the organization named “3rd Generation Partnership Project” (3GPP). CDMA2000 and UMB are described in documents from an organization named “3rd Generation Partnership Project 2” (3GPP2). The techniques described herein may be used for the systems and radio technologies mentioned herein as well as other systems and radio technologies. While aspects of an LTE, LTE-A, LTE-A Pro, or NR system may be described for purposes of example, and LTE, LTE-A, LTE-A Pro, or NR terminology may be used in much of the description, the techniques described herein are applicable beyond LTE, LTE-A, LTE-A Pro, or NR applications.

(243) A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs with service subscriptions with the network provider. A small cell may be associated with a lower-powered base station, as compared with a macro cell, and a small cell may operate in the same or different (e.g., licensed, unlicensed, etc.) frequency bands as macro cells. Small cells may include pico cells, femto cells, and micro cells according to various examples. A pico cell, for example, may cover a small geographic area and may allow unrestricted access by UEs with service subscriptions with the network provider. A femto cell may also cover a small geographic area (e.g., a home) and may provide restricted access by UEs having an association with the femto cell (e.g., UEs in a closed subscriber group (CSG), UEs for users in the home, and the like). An eNB for a macro cell may be referred to as a macro eNB. An eNB for a small cell may be referred to as a small cell eNB, a pico eNB, a femto eNB, or a home eNB. An eNB may support one or multiple (e.g., two, three, four, and the like) cells, and may also support communications using one or multiple component carriers.

(244) The wireless communications systems described herein may support synchronous or asynchronous operation. For synchronous operation, the base stations may have similar frame timing, and transmissions from different base stations may be approximately aligned in time. For asynchronous operation, the base stations may have different frame timing, and transmissions from different base stations may not be aligned in time. The techniques described herein may be used for either synchronous or asynchronous operations.

(245) Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(246) The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA, or other programmable logic device, discrete gate or transistor logic, discrete hardware

components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(247) The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

(248) Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, non-transitory computer-readable media may include random-access memory (RAM), read-only memory (ROM), electrically erasable programmable ROM (EEPROM), flash memory, compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

(249) As used herein, including in the claims, “or” as used in a list of items (e.g., a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary feature that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

(250) In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label, or other subsequent reference label.

(251) The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an

example, instance, or illustration," and not "preferred" or "advantageous over other examples." The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.
(252) The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. An apparatus for wireless communications at a first wireless device, comprising: one or more memories; and one or more processors coupled with the one or more memories and configured to cause the first wireless device to: determine a speed of the first wireless device; identify a subcarrier spacing for communications with one or more other wireless devices; determine, based at least in part on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulation of data; transmit an indication of the reference signal pattern to the one or more other wireless devices; and transmit the data and the reference signal to the one or more other wireless devices, wherein the reference signal is transmitted in accordance with the reference signal pattern.
2. The apparatus of claim 1, wherein, to transmit the data and the reference signal, the one or more processors are configured to cause the first wireless device to: broadcast the data and the reference signal to the one or more other wireless devices.
3. The apparatus of claim 1, wherein the indication of the reference signal pattern is transmitted via a sidelink control information message.
4. The apparatus of claim 1, wherein, to determine the reference signal pattern of the reference signal, the one or more processors are configured to cause the first wireless device to: select the reference signal pattern from one or more reference signal patterns, each of the one or more reference signal patterns corresponding to a respective speed, a range of speeds, or a combination thereof.
5. The apparatus of claim 1, wherein the one or more processors are configured to cause the first wireless device to: determine a direction of the first wireless device, wherein the reference signal pattern is based at least in part on the direction.
6. The apparatus of claim 1, wherein the one or more processors are configured to cause the first wireless device to: determine that the first wireless device is unable to receive assistance information from the one or more other wireless devices, wherein the speed, the subcarrier spacing, the reference signal pattern, or any combination thereof, is determined based on the first wireless device being unable to receive assistance information from the one or more other wireless devices.
7. The apparatus of claim 6, wherein the one or more processors are configured to cause the first wireless device to: estimate an additional speed, a direction, or both, of the one or more other wireless devices based at least in part on the first wireless device being unable to receive assistance information from the one or more other wireless devices, wherein the reference signal pattern is based at least in part on the additional speed, the direction, or both.
8. The apparatus of claim 1, wherein the reference signal pattern comprises a demodulation reference signal pattern, and wherein the reference signal comprises a demodulation reference signal.
9. A method for wireless communications at a first wireless device, comprising: determining a

speed of the first wireless device; identifying a subcarrier spacing for communications with one or more other wireless devices; determining, based at least in part on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulating data; transmitting an indication of the reference signal pattern to the one or more other wireless devices; and transmitting the data and the reference signal to the one or more other wireless devices, wherein the reference signal is transmitted in accordance with the reference signal pattern.
10. The method of claim 9, wherein transmitting the data and the reference signal comprises: broadcasting the data and the reference signal to the one or more other wireless devices.
11. The method of claim 9, wherein the indication of the reference signal pattern is transmitted via a sidelink control information message.
12. The method of claim 9, wherein determining the reference signal pattern of the reference signal comprises: selecting the reference signal pattern from one or more reference signal patterns, each of the one or more reference signal patterns corresponding to a respective speed, a range of speeds, or a combination thereof.
13. The method of claim 9, further comprising: determining a direction of the first wireless device, wherein the reference signal pattern is based at least in part on the direction.
14. The method of claim 9, further comprising: determining that the first wireless device is unable to receive assistance information from the one or more other wireless devices, wherein determining the speed, identifying the subcarrier spacing, determining the reference signal pattern, or any combination thereof, is based on the first wireless device being unable to receive assistance information from the one or more other wireless devices.
15. The method of claim 14, further comprising: estimating an additional speed, a direction, or both, of the one or more other wireless devices based at least in part on the first wireless device being unable to receive assistance information from the one or more other wireless devices, wherein the reference signal pattern is based at least in part on the additional speed, the direction, or both.
16. The method of claim 9, wherein the reference signal pattern comprises a demodulation reference signal pattern, and wherein the reference signal comprises a demodulation reference signal.
17. A non-transitory computer-readable medium storing code for wireless communication at a first wireless device, the code comprising instructions executable by one or more processors to cause the first wireless device to: determine a speed of the first wireless device; identify a subcarrier spacing for communications with one or more other wireless devices; determine, based at least in part on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulation of data; transmit an indication of the reference signal pattern to the one or more other wireless devices; and transmit the data and the reference signal to the one or more other wireless devices, wherein the reference signal is transmitted in accordance with the reference signal pattern.
18. The non-transitory computer-readable medium of claim 17, wherein, to transmit the data and the reference signal, the instructions are further executable by the one or more processors to cause the first wireless device to: broadcast the data and the reference signal to the one or more other wireless devices.
19. The non-transitory computer-readable medium of claim 17, wherein, to determine the reference signal pattern of the reference signal, the instructions are further executable by the one or more processors to cause the first wireless device to: select the reference signal pattern from one or more reference signal patterns, each of the one or more reference signal patterns corresponding to a respective speed, a range of speeds, or a combination thereof.
20. The non-transitory computer-readable medium of claim 17, wherein the instructions are further executable by the one or more processors to cause the first wireless device to: determine a direction of the first wireless device, wherein the reference signal pattern is based at least in part on the

direction.

21. The non-transitory computer-readable medium of claim 17, wherein the reference signal pattern comprises a demodulation reference signal pattern, and wherein the reference signal comprises a demodulation reference signal.

22. An apparatus for wireless communication at a first wireless device, comprising: means for determining a speed of the first wireless device; means for identifying a subcarrier spacing for communications with one or more other wireless devices; means for determining, based at least in part on the speed of the first wireless device and the subcarrier spacing, a reference signal pattern of a reference signal for demodulation of data; means for transmitting an indication of the reference signal pattern to the one or more other wireless devices; and means for transmitting the data and the reference signal to the one or more other wireless devices, wherein the reference signal is transmitted in accordance with the reference signal pattern.

23. The apparatus of claim 22, wherein the means for transmitting the data and the reference signal comprise: means for broadcasting the data and the reference signal to the one or more other wireless devices.

24. The apparatus of claim 22, wherein the indication of the reference signal pattern is transmitted via a sidelink control information message.

25. The apparatus of claim 22, wherein the reference signal pattern comprises a demodulation reference signal pattern, and wherein the reference signal comprises a demodulation reference signal.